

**An AI-Driven Smart Platform for Reducing Waste and
Improving Efficiency in Sri Lanka's Fruit
Supply Chain**

25-26J-444

Draft Thesis Report

B.Sc. (Hons) Degree in Information Technology Specializing in
Software Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

April 2026

DECLARATION

We declare that this dissertation is our own work and that it does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any other university or institute of higher learning. To the best of our knowledge and belief, it does not contain any material previously published or written by another person except where due acknowledgement is made in the text. We hereby grant to the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this dissertation, in whole or in part, in print, electronic, or other medium. We retain the right to use this content in whole or in part in future works such as journal articles or books.

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ABSTRACT

Sri Lanka's fruit supply chain is affected by post-harvest losses, unstable market information, delayed payment, quality disputes, weak logistics coordination, and high dependency on intermediaries. This group thesis presents FreshRoute, a smart fruit supply chain platform designed to reduce waste and support direct, trustworthy trade between smallholder farmers and buyers. The system combines four coordinated components: an AI-driven demand forecasting and alert service, a blockchain-powered direct marketplace with verifiable reputation, a spoilage-aware logistics and telemetry engine, and an explainable multi-attribute fruit quality grading module. The methodology follows a design science research approach supported by iterative software engineering. Requirements were derived from literature, stakeholder needs, and the team's individual thesis artifacts, then translated into an integrated architecture using Node.js/Express services, PostgreSQL/Supabase storage, Hyperledger Fabric concepts, FastAPI forecasting services, MobileNetV3-based image grading, routing services, notification gateways, and IoT telemetry. Evaluation combines prototype inspection, module-level test planning, usability criteria, and the available logistics benchmark evidence. The benchmarked logistics component eliminated unsafe vehicle allocations in the tested synthetic scenarios while maintaining slightly higher fulfillment than random and naive baselines, although with increased runtime. The study concludes that FreshRoute is a credible proof-of-concept for a Sri Lankan digital fruit supply chain platform. Field pilots, stronger security hardening, production deployment practices, and larger real datasets are required before national-scale adoption.

Keywords: FreshRoute, fruit supply chain, post-harvest loss, demand forecasting, blockchain marketplace, logistics optimization, fruit quality grading

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We also extend our appreciation to our co-supervisor, Ms. Poojani Gunathilake, for her valuable support, suggestions, and technical advice. Her feedback helped the team balance research ambition with practical implementation constraints and improve the clarity of the system design.

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Finally, we thank our families, friends, and team members for their patience, motivation, and collaboration. FreshRoute is a group effort, and the integration of forecasting, marketplace, logistics, and quality assurance was possible only through continuous coordination and shared responsibility.

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LIST OF ABBREVIATIONS

Table C: Abbreviations used in the thesis.

Abbreviation	Full form
AI	Artificial Intelligence
API	Application Programming Interface
ARIMA	Autoregressive Integrated Moving Average
B2B	Business-to-Business
CA	Certificate Authority
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
DID	Decentralized Identifier
DSR	Design Science Research
ETA	Estimated Time of Arrival
FCM	Firestore Cloud Messaging
GPS	Global Positioning System
HARTI	Hector Kobbekaduwa Agrarian Research and Training Institute
HTTPS	Hypertext Transfer Protocol Secure
IoT	Internet of Things
IPFS	InterPlanetary File System
JSON	JavaScript Object Notation
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MSE	Mean Squared Error
ONNX	Open Neural Network Exchange
OSRM	Open Source Routing Machine
PHL	Post-Harvest Loss
REST	Representational State Transfer
RMSE	Root Mean Square Error
SARIMA	Seasonal Autoregressive Integrated Moving Average
SLIIT	Sri Lanka Institute of Information Technology
SMS	Short Message Service
SQL	Structured Query Language
TLS	Transport Layer Security
UAT	User Acceptance Testing
UI/UX	User Interface / User Experience
VC	Verifiable Credential
VRP	Vehicle Routing Problem
WAPE	Weighted Absolute Percentage Error

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1 INTRODUCTION

1.1 Background and literature survey

Sri Lanka has favorable agro-ecological conditions for fruit cultivation, but the economic and social value of fruit production is reduced by persistent inefficiencies after harvest. The supply chain is highly fragmented. Smallholder farmers often sell through village collectors, commission agents, wholesalers, transport providers, and retailers before produce reaches consumers or institutional buyers. Each hand-off increases the possibility of mechanical damage, delay, price distortion, and information loss. When fruits such as banana, mango, papaya, pineapple, and guava move through poorly coordinated channels, quality deterioration begins before the final buyer can fully assess the product. National and commodity-level studies have repeatedly identified poor handling, inadequate packaging, limited cold-chain availability, and transport stress as important causes of post-harvest loss in the Sri Lankan fruit and vegetable sector [1], [2], [4], [6].

The problem is not simply that fruit is physically damaged. Post-harvest loss also reduces farmer income, raises consumer prices, wastes land and water, and weakens trust among producers, buyers, and consumers. Farmers face a short decision window because ripe fruit cannot be held indefinitely. When they do not know expected demand, current buyer requirements, or regional price movement, they may harvest at the same time as neighboring producers and create oversupply. Oversupply depresses prices, while delayed sale or unsuitable transport increases spoilage. The economic burden is severe because farmers bear risk at the earliest and most vulnerable point of the chain, while actors closer to urban markets often possess stronger price information and bargaining power.

Intermediaries play a complex role in this environment. They provide aggregation, market access, informal credit, and transport coordination, and these functions are often valuable for small farmers who cannot independently reach institutional buyers. However, dependency on intermediaries also creates information asymmetry. The farmer may not know the buyer's actual willingness to pay, quality requirement, or final retail price. The buyer may not know the farmer's reliability, harvest history, or

product quality. This dual trust gap explains why simply publishing a digital listing is not enough. A sustainable alternative must reproduce the useful coordination functions of intermediaries while reducing opacity, unfair pricing, delayed settlement, and accountability gaps.

FreshRoute was designed as a response to this socio-technical problem. It is not a single-purpose mobile application. It is a smart fruit supply chain platform that connects demand intelligence, direct marketplace transactions, quality verification, spoilage-aware transport, payments, notifications, and traceability. The group project contains four mutually reinforcing components. The demand forecasting module helps farmers plan harvesting and listing decisions before fruit enters the market. The blockchain marketplace module supports direct farmer-buyer trade, reputation, and settlement confidence. The logistics module allocates vehicles and monitors transport conditions based on perishability. The quality grading module applies image-based evaluation to improve fairness in grading and complaint handling. Figure 1.1 summarizes the conventional chain that motivates the intervention, while Figure 1.2 illustrates the integrated FreshRoute response.

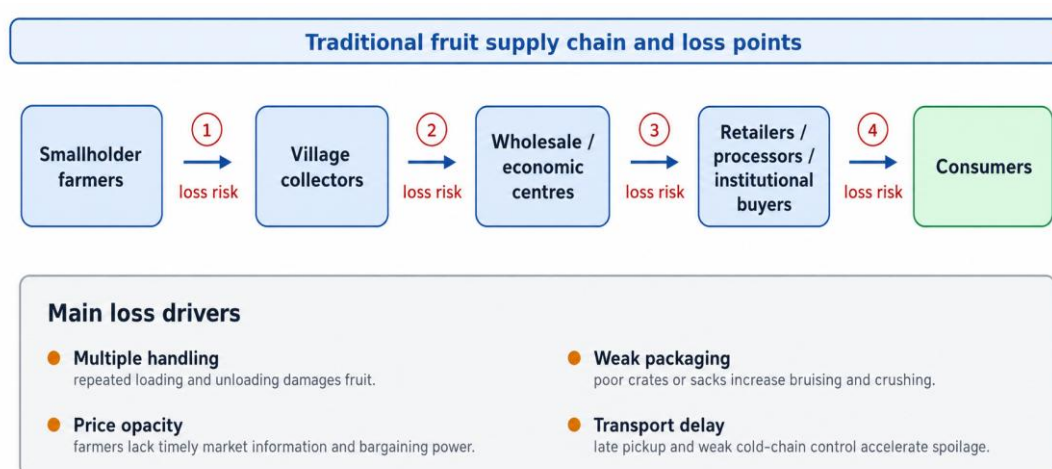


Figure 1.1: Traditional fruit supply chain and loss points.

Source: Prepared by the FreshRoute project team based on Sri Lankan supply-chain literature [1]-[9].

The literature on Sri Lankan post-harvest loss shows that technical and institutional causes reinforce each other. Rajapaksha et al. describe how harvesting maturity, packaging, transportation, storage, and market practices combine to produce fruit and

vegetable losses [1]. Banana loss studies show that downstream transport and handling stages can contribute materially to both loss volume and environmental impact [2], [6]. Mango studies similarly show that improved post-harvest practices can extend shelf life and reduce loss when they are applied consistently [5]. These studies are important for FreshRoute because they show that digital technology must connect operational decisions to real handling conditions. If a platform only records orders but leaves vehicle choice and timing unmanaged, it does not directly address one of the most loss-prone stages.

A second body of literature focuses on value chain structure, price distribution, and intermediary control. HARTI and related studies show that fruit and vegetable chains in Sri Lanka contain several actors between the producer and final consumer [3], [8], [25]. Such chains may be necessary for aggregation, but they often create low visibility for farmers and buyers. Research on intermediary margins and post-harvest waste indicates that the cost of losses and commissions can be embedded in final prices [24]. For FreshRoute, this means that the platform must not treat middlemen only as an abstract social problem. It must provide concrete alternatives for the activities that intermediaries currently perform: trust creation, aggregation, transport coordination, payment assurance, and quality dispute handling.

A third body of research concerns digital agriculture and forecast-driven decision support. Traditional time-series models such as ARIMA and SARIMA remain useful when agricultural prices have stable trends and seasonal patterns, while Prophet is a strong interpretable baseline for trend and seasonality [10]. Recurrent neural networks, especially Long Short-Term Memory models, are often more suitable for nonlinear agricultural demand because they can learn dependencies across weather patterns, prior price movement, holidays, and delayed supply effects [11], [12]. However, forecasting only creates value when results reach farmers in time to influence behavior. This makes mobile dashboards, SMS fallback, offline caching, and understandable advice as important as model accuracy [20], [21].

A fourth research stream concerns blockchain, traceability, decentralized identity, and smart contracts. Permissioned blockchain platforms such as Hyperledger Fabric are well suited to business-to-business supply chains because participants are

identifiable and commercial data may require privacy [16]. International agri-food blockchain studies show that distributed ledgers can improve transparency, traceability, and trust when they are integrated with actual operational workflows [17]. For Sri Lankan smallholders, the key requirement is not public speculation on cryptocurrency but controlled, verifiable evidence of identity, deliveries, quality outcomes, and settlement. Decentralized Identifiers and Verifiable Credentials are therefore relevant because they can convert repeated successful transactions into portable digital reputation.

A fifth research stream relates to logistics optimization and perishable product routing. Traditional vehicle routing focuses on distance, capacity, and cost, but fruits need additional rules because thermal exposure, fragility, ripeness, rain, and journey duration affect quality. Osvald and Stirn demonstrate that perishable goods routing can include freshness-sensitive constraints [18], while later reviews confirm that vehicle routing for perishable goods needs to balance travel time, product deterioration, vehicle capacity, and service windows [19]. FreshRoute adopts this insight by treating vehicle allocation as a freshness decision, not merely a transport cost decision. The logistics module uses rules for temperature tolerance, fruit compatibility, distance, and telemetry alerts to protect consignments during delivery.

A final stream concerns image-based quality grading. Classical computer vision approaches use color histograms, texture descriptors, and hand-crafted features, while modern deep learning approaches use convolutional neural networks and transfer learning [13]. MobileNetV3 is relevant for a student prototype because it balances accuracy with computational efficiency for future mobile or lightweight deployment [14]. Explainability methods such as Grad-CAM provide visual evidence of which image regions influenced a prediction [15]. FreshRoute uses these ideas in the quality component by moving beyond a black-box grade-only classifier toward a multi-attribute grading design that predicts color stage, spot severity, blemish severity, and final grade.

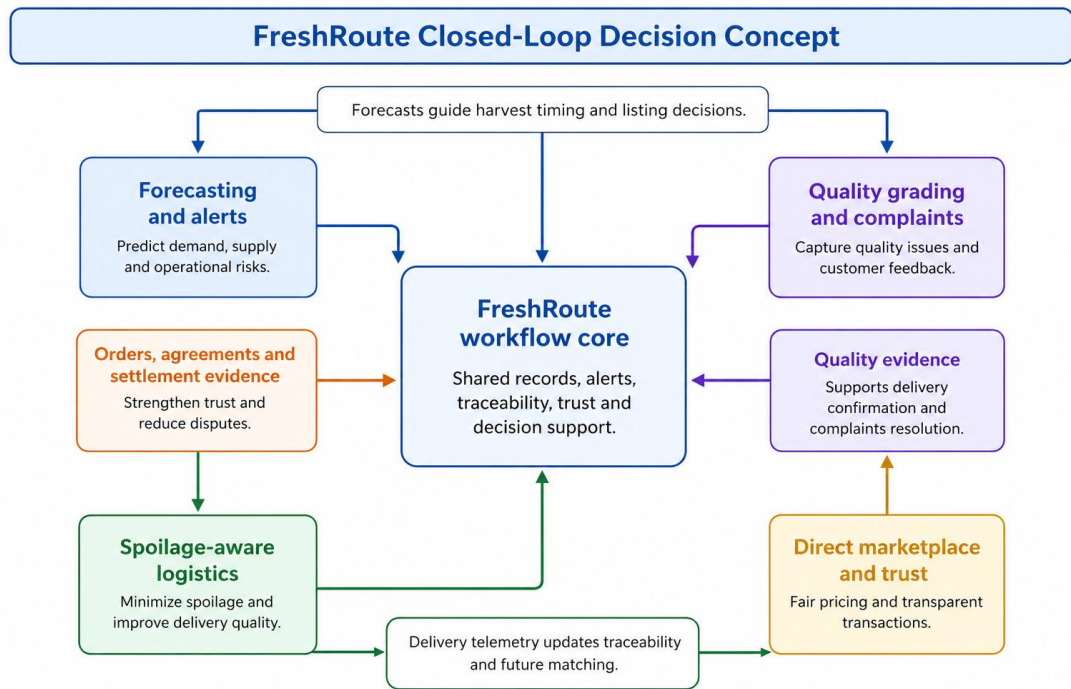


Figure 1.2: FreshRoute closed-loop decision concept.

Source: Prepared by the FreshRoute project team from the four individual component designs.

Table 1.1: Summary of selected literature relevant to FreshRoute.

Research area	Representative sources	Insight for FreshRoute
Post-harvest loss in Sri Lanka	[1], [2], [4], [5], [6]	Loss is driven by handling, maturity, packaging, transportation, and cold-chain weakness.
Value chain and intermediary structure	[3], [7], [8], [24], [25]	Multiple actors provide useful functions but create price opacity, dependency, and accountability gaps.
Demand and price forecasting	[10], [11], [12], [20], [21]	Forecasts become useful only when converted into simple, timely, and accessible farmer advisories.
Blockchain and trust	[16], [17], [29]	Permissioned ledgers and verifiable credentials can support trustworthy B2B trade and reputation.
Perishable logistics	[18], [19], [28]	Routing and vehicle assignment should consider freshness risk, temperature, and service windows.
Fruit grading and explainable AI	[13], [14], [15]	Image grading should explain visible cues such as color, spots, blemishes, and final quality grade.

Source: Prepared by the FreshRoute project team from the reviewed sources.

1.2 Research gap

The literature shows that individual solutions exist for parts of the problem, but the Sri Lankan fruit supply chain still lacks an integrated operational platform that connects prediction, trade, trust, transport, and quality in one workflow. Training programs and packaging interventions can reduce handling losses, but they do not

automatically decide when a farmer should harvest, which buyer should receive the produce, what vehicle should collect it, or how a quality dispute should be resolved. Forecasting studies can produce useful models, but many remain laboratory experiments without rural alert delivery, marketplace integration, or logistics consequences. Marketplace platforms can list produce, but they often do not solve payment confidence, reputation, aggregation, or product traceability.

This gap is especially important because the supply chain problems occur together. If demand is forecast accurately but the farmer still sells through an opaque channel, the forecast does not improve bargaining power. If a buyer places a direct order but cannot verify farmer reliability, the buyer may return to intermediaries. If a direct order is created but the fruit is moved in an unsuitable vehicle, spoilage remains high. If quality is disputed after delivery but no image evidence or traceability record exists, trust collapses. FreshRoute therefore addresses an integration gap rather than a single algorithmic gap.

There is also a contextual gap. Many international agri-tech solutions assume stronger infrastructure, higher smartphone adoption, larger producers, and better cold-chain availability. Sri Lankan smallholders require designs that tolerate intermittent connectivity, limited digital literacy, fragmented production, smaller loads, and variable vehicle capacity. The platform must therefore combine advanced techniques with simple user-facing outputs: advisory messages, verified orders, transparent payment states, practical vehicle recommendations, and clear quality explanations.

Table 1.2: Research gap comparison matrix.

Capability	Typical information app	Marketplace-only app	Traceability-only system	FreshRoute group platform
Predictive harvest and price support	Partial	No	No	Yes, through AI forecasting and alerts
Direct farmer-buyer trade	No	Yes	Partial	Yes, with matching and aggregation
Verifiable reputation and settlement evidence	No	Limited	Partial	Yes, through DID/VC and smart agreement logic
Spoilage-aware vehicle assignment	No	No	No	Yes, through logistics rules and routing
IoT and delivery condition alerts	No	No	Partial	Yes, through telemetry and notification services
Explainable fruit quality grading	No	Partial	Partial	Yes, through multi-attribute grading
Unified system architecture	No	Partial	Partial	Yes, through integrated API and shared data flows

Source: Prepared by the FreshRoute project team based on individual component reports [34], [35].

1.3 Research problem

The research problem addressed by FreshRoute is that Sri Lanka's fruit supply chain lacks an accessible and integrated digital mechanism that enables smallholder farmers and buyers to coordinate harvest planning, direct trade, quality assurance, payment confidence, and freshness-aware logistics without excessive dependence on opaque intermediaries. Current practices expose farmers to delayed information, low bargaining power, distress sales, and post-harvest loss. Buyers also face risk because fragmented smallholder supply can be difficult to verify, aggregate, transport, and grade consistently.

The problem has four interconnected technical dimensions. First, farmers need timely forecasts and actionable alerts before making harvest decisions. Second, direct trade requires verified identity, reliable reputation, aggregation, and settlement confidence. Third, perishable fruit delivery requires vehicle selection and route planning that reflect fruit sensitivity and environmental risk. Fourth, quality disputes require objective and explainable evidence rather than only subjective visual inspection after a problem occurs. Solving only one dimension leaves the remaining chain vulnerable.

The practical research problem can therefore be stated as follows: How can a smart fruit supply chain platform be designed and evaluated to reduce post-harvest waste and intermediary dependency in Sri Lanka by integrating AI-based demand forecasting, blockchain-based marketplace trust, spoilage-aware logistics, and explainable fruit quality grading into one coordinated software artifact?

1.4 Research questions

1. How can a multi-module digital platform support Sri Lankan fruit farmers and buyers by reducing information asymmetry, improving direct market access, and increasing operational transparency?
2. How can demand forecasting and alert delivery translate market, weather, and seasonal data into actionable harvest-planning guidance for smallholder farmers?

3. How can a blockchain-based marketplace with verifiable identity, reputation, aggregation, and smart sales agreement logic reduce trust barriers in direct farmer-buyer trade?
4. How can spoilage-aware logistics combine vehicle suitability, routing, weather exposure, and telemetry to protect fruit freshness during distribution?
5. How can image-based fruit quality grading become more explainable through visible attributes such as color stage, spot severity, blemish severity, and final grade?
6. What prototype-level evidence, limitations, and future work remain before FreshRoute can be piloted at production scale?

1.5 Research objectives

The main objective of this group thesis is to design, develop, and evaluate FreshRoute as an integrated smart fruit supply chain platform for reducing waste and dependency on intermediaries in Sri Lanka. The platform aims to coordinate harvest planning, transparent trade, perishability-aware logistics, quality verification, payment visibility, notifications, and traceability in a single ecosystem.

The specific objectives are organized around the four student components and the integrated architecture. The first objective is to design an AI-driven demand forecasting and alert service that uses market, weather, and seasonal data to generate farmer-facing advisories. The second objective is to design a direct marketplace and trust layer that uses verified onboarding, decentralized identity concepts, verifiable reputation, smart agreement logic, and aggregation. The third objective is to design a logistics module that assigns appropriate vehicles, calculates executable routes, and monitors delivery conditions for perishable fruit loads. The fourth objective is to design an explainable quality grading module that predicts visible quality attributes and final fruit grade from images. The fifth objective is to evaluate the prototype through technical review, test planning, simulation evidence, and discussion of deployment readiness.

Table 1.3: Alignment between research questions and objectives.

Research question	Primary objective	Expected evidence
RQ1: Integrated platform support	Define the FreshRoute architecture and shared workflow	Architecture diagrams, stakeholder analysis, component integration evidence

RQ2: Demand forecasting	Develop forecasting and alert methodology	Data pipeline, model design, API and alert workflow, evaluation metrics
RQ3: Marketplace trust	Develop direct trading and reputation logic	DID/VC model, smart agreement lifecycle, matching and aggregation design
RQ4: Logistics freshness	Develop spoilage-aware transport coordination	Vehicle rules, routing workflow, telemetry design, benchmark evidence
RQ5: Explainable quality grading	Develop multi-attribute grading methodology	Annotation rubric, MobileNetV3 design, rule-based and Grad-CAM explanation plan
RQ6: Prototype readiness	Evaluate evidence and identify limitations	Module result matrix, benchmark results, risks, recommendations

Source: Prepared by the FreshRoute project team.

1.6 Scope and delimitations

The scope of this group thesis covers the FreshRoute prototype as a final-year software engineering research artifact. It includes demand forecasting and alert design, blockchain marketplace design, logistics assignment and telemetry design, quality grading design, shared architecture, commercialization considerations, and testing strategy. The platform is framed around selected high-priority fruits such as banana, mango, and papaya, with Sri Lankan market centers and producing districts treated as the intended operational context.

The study is delimited in several ways. It does not claim national deployment impact because no long-term field pilot across real farmers, buyers, fleets, and markets has been completed. It does not claim laboratory freshness measurement because the available logistics evidence is based on rule-based benchmark and simulation outputs. It does not claim final production security because additional penetration testing, key management, privacy audits, and regulated payment integration are required. It also does not claim that all intermediaries should disappear; rather, the goal is to reduce dependency on opaque intermediaries by digitizing useful coordination functions in a more transparent way.

These delimitations are important because they keep the thesis academically honest. FreshRoute demonstrates a coherent design and a credible proof-of-concept, but it remains a prototype that requires data expansion, pilot validation, user training, and production hardening. The contribution is therefore best understood as a design-science artifact and a practical model for future field deployment, not as a finished national infrastructure system.

Table 1.4: Scope and delimitations.

Included in scope	Excluded or partially covered
-------------------	-------------------------------

Integrated architecture for forecasting, marketplace, logistics, and quality modules	Full national deployment across all fruit supply chains
Prototype design, backend/service flows, data model, and evaluation strategy	Long-term field trials with statistically representative farmers and buyers
Synthetic logistics benchmark and scenario analysis	Laboratory measurement of biochemical fruit deterioration
Demand forecasting methodology and alert design	Guaranteed model performance on all crops without final production datasets
Blockchain marketplace and reputation framework	Regulated real-money escrow deployment without banking/payment compliance review
Image-based external quality grading	Internal fruit quality, sweetness, or destructive testing

Source: Prepared by the FreshRoute project team.

1.7 Organization of the thesis

Chapter 1 introduces the research context, literature, gap, problem, questions, objectives, and scope. Chapter 2 presents the methodology, including research design, stakeholder analysis, system architecture, module methodologies, algorithms, tools, commercialization, implementation, testing, and ethical considerations. Chapter 3 presents results and discussion, including prototype evidence, module evaluation, research findings, limitations, and each student's contribution. Chapter 4 provides conclusions, recommendations, and future work. The reference list, glossary, and appendices provide supporting bibliographic, definitional, requirement, testing, budget, risk, and pilot-planning material.

1.8 Chapter summary

This chapter established the need for FreshRoute by showing that Sri Lankan fruit supply-chain waste and intermediary dependency are caused by a combination of physical, informational, economic, and trust-related weaknesses. It reviewed literature relevant to post-harvest loss, forecasting, blockchain trust, perishable logistics, and image-based grading. The chapter then defined the integrated research gap and framed the group thesis around a multi-module software artifact. The next chapter explains how the platform was designed, implemented, and prepared for evaluation.

2 METHODOLOGY

2.1 Research approach and design

FreshRoute follows a design science research approach because the main contribution is a purposeful digital artifact designed to address a real socio-technical problem. Design science research connects problem identification, objective definition, artifact construction, demonstration, evaluation, and communication [22], [23]. This is appropriate for FreshRoute because the project does not only describe inefficiencies in the fruit supply chain; it designs a platform that can reduce those inefficiencies through coordinated software services.

The team also adopted an iterative software engineering approach. A purely linear development method would be unsuitable because requirements, data availability, stakeholder priorities, and module interfaces evolved during the final-year project. The work was therefore organized into problem analysis, requirement elicitation, architecture design, module implementation, integration, testing, and evaluation planning. Each component progressed through prototype cycles while the group repeatedly reviewed whether forecasting, marketplace, logistics, and quality outputs could exchange usable data.

The research design combines four evidence sources. First, literature and institutional reports were reviewed to understand post-harvest loss, supply-chain fragmentation, digital agriculture, blockchain traceability, routing, and fruit grading. Second, the team's individual thesis documents were analyzed to consolidate component-specific requirements and methods. Third, the final project artifact and service structure were reviewed to infer the implemented architecture and integration points. Fourth, available benchmark and simulation outputs were interpreted to discuss prototype-level results, especially for the logistics module.

2.2 Requirement gathering and stakeholder analysis

Requirements were derived from the needs of four primary user groups: farmers, buyers, transporters, and administrators. Farmers require timely market intelligence, a fair route to buyers, trusted payment, simple listing workflows, and visibility of quality and delivery status. Buyers require reliable supply, verified farmer reputation,

quality evidence, order aggregation, delivery tracking, and settlement documentation. Transporters require clear pickup and drop instructions, feasible routes, vehicle-fit decisions, and telemetry-based alerts. Administrators require governance functions, dispute handling, user verification, system monitoring, and reporting.

The project also considered institutional partners such as agricultural agencies, quality assessors, payment providers, and market centers. These actors may not all use the platform daily, but they influence adoption. For example, an agricultural officer could support farmer onboarding and training; a buyer organization could define quality standards; a payment partner could provide compliant settlement; a local logistics provider could operate reefer and non-reefer vehicles. FreshRoute must therefore be modular enough to integrate with existing institutions rather than assuming that technology alone will reorganize the entire market.

Requirement gathering converted stakeholder needs into functional and non-functional requirements. Functional requirements define what the system should do: generate forecasts, list produce, match farmers and buyers, create agreements, allocate vehicles, monitor telemetry, grade images, process payments, and record evidence. Non-functional requirements define how the system should behave: secure, scalable, reliable, explainable, accessible, maintainable, and suitable for rural environments. Tables 2.1 to 2.3 summarize the stakeholder roles and major requirements.

Table 2.1: Stakeholder roles and major needs.

Stakeholder	Role in FreshRoute	Major needs
Farmer	Lists produce, views forecasts, accepts matches, receives alerts, and builds reputation.	Simple UI, language clarity, offline access, fair pricing, payment confidence, quality transparency.
Buyer	Places orders, reviews matches, confirms quality and settlement, tracks delivery.	Verified suppliers, aggregation, traceability, quality assurance, transparent payment and delivery status.
Transporter	Executes assigned pickups and deliveries and transmits telemetry.	Feasible jobs, route guidance, vehicle-fit logic, shift awareness, GPS/temperature alerts.
Administrator	Manages users, governance, disputes, data quality, and system health.	Audit logs, dashboards, security controls, exception handling, report exports.
External partner	Provides weather, routing, payment, SMS, blockchain, or institutional support.	Clear APIs, compliance, service-level reliability, privacy controls.

Source: Prepared by the FreshRoute project team.

Table 2.2: Core functional requirements of FreshRoute.

ID	Requirement	Description
FR1	Forecast generation	The platform shall generate fruit demand

		and price forecasts using historical market, weather, and seasonal data.
FR2	Alert delivery	The platform shall convert forecasts into farmer-friendly advisories and dispatch mobile or SMS notifications.
FR3	Verified onboarding	The platform shall register farmers, buyers, transporters, and administrators with role-based access.
FR4	Supply listing and order placement	Farmers shall list produce and buyers shall create demand records with quantity, grade, date, and location.
FR5	Matching and aggregation	The system shall rank and combine farmer listings to satisfy buyer requirements.
FR6	Smart agreement lifecycle	The system shall record order acceptance, escrow status, delivery confirmation, quality verification, and settlement events.
FR7	Vehicle assignment and routing	The system shall allocate vehicles based on fruit sensitivity, capacity, route distance, weather, and driver constraints.
FR8	Telemetry and alerts	The transporter workflow shall support GPS, temperature, humidity, and event logs for delivery monitoring.
FR9	Quality grading	The platform shall support image-based grading with visible attributes and final grade prediction.
FR10	Traceability and reporting	Authorized users shall view transaction, delivery, quality, and settlement evidence.

Source: Prepared by the FreshRoute project team from individual component reports [34], [35].

Table 2.3: Non-functional requirements.

Requirement	Meaning for FreshRoute
Reliability	Core workflows should return consistent results and preserve transaction and delivery evidence.
Performance	Forecast retrieval, matching, and vehicle allocation should remain responsive under prototype-scale load.
Security	Role-based access, TLS/HTTPS, certificate-based trust, safe payment handling, and data privacy controls are required.
Scalability	The architecture should support additional crops, districts, buyers, and logistics partners.
Accessibility	Farmer-facing workflows should use simple language, offline-friendly data, and SMS fallback.
Explainability	Forecast confidence, vehicle suitability reasons, reputation evidence, and quality attributes should be understandable.
Maintainability	The platform should separate controllers, services, models, data access, and external integrations.
Compliance	Pilot deployment should consider payment regulation, data protection, consent, and ethical handling of stakeholder data.

Source: Prepared by the FreshRoute project team.

2.3 Overall system architecture

The FreshRoute architecture is designed as a modular service-oriented platform. Users interact through farmer, buyer, transporter, and administrator interfaces. These interfaces communicate with an API layer that validates requests and routes them to the relevant services. The business services include forecasting, marketplace matching, logistics, quality grading, payment, notification, and traceability functions. A shared data layer stores users, listings, orders, forecasts, vehicles, trips, telemetry, quality records, complaints, and settlement status. External services provide weather information, routing matrices, SMS, push notifications, payment integration, and blockchain gateway functions.

This architecture supports separation of concerns. Forecasting can evolve from a baseline model to an LSTM or hybrid model without rewriting marketplace logic. The marketplace can improve reputation scoring without altering image preprocessing. Logistics can switch between routing providers or vehicle allocation heuristics while retaining the order data model. Quality grading can export ONNX or TorchScript models for deployment while the API continues to expose a stable quality endpoint. This modularity is critical because final-year prototypes often evolve unevenly across components, and a clean boundary reduces integration risk.

The architecture also supports a data-driven operational loop. Forecasts influence farmer listing timing. Listings and buyer orders produce matches. Matches create agreements and transport requirements. Logistics events and telemetry create delivery evidence. Quality grading and complaint validation create quality evidence. Successful transaction outcomes update reputation. Updated reputation improves future matching. This loop is the core systems idea behind FreshRoute: each module improves the usefulness of the others by producing data that can be reused downstream.

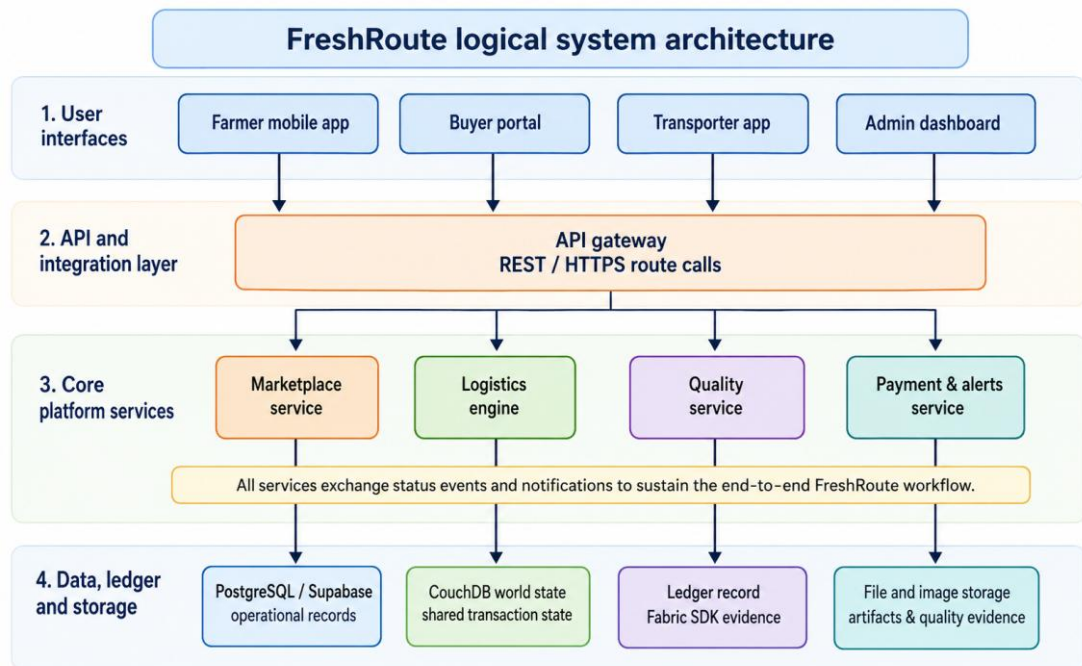


Figure 2.1: FreshRoute logical system architecture.

Source: Prepared by the FreshRoute project team from the consolidated component architecture.

Table 2.4: Module responsibilities and artifact evidence.

Module	Primary responsibility	Evidence in component documents
Demand forecasting	Predict demand and price direction and generate farmer advisories.	LSTM methodology, FastAPI service design, PostgreSQL/Redis storage, SMS/FCM alerts.
Direct marketplace	Create trusted farmer-buyer trade with identity, reputation, matching, and agreement logic.	Hyperledger Fabric design, DID/VC reputation, smart sales agreement, aggregation algorithm.
Logistics	Assign vehicles, generate routes, and monitor delivery conditions for perishable fruit.	Vehicle suitability rules, OSRM/weather integration, telemetry gateway, benchmark and Matara simulation evidence.
Quality grading	Grade fruit images and provide visible explanation for quality decisions.	MobileNetV3-large backbone, attribute labels, Grad-CAM plan, direct and rule-based grade comparison.
Shared services	Coordinate payments, notifications, authentication, logs, and admin governance.	Node.js/Express route organization, PayHere/Twilio/Firebase concepts, database and ledger evidence flows.

Source: Prepared by the FreshRoute project team from uploaded component artifacts [34], [35].

2.4 Demand forecasting and alert methodology

The demand forecasting component forms the intelligence layer of FreshRoute. Its purpose is to support harvest planning before produce enters the market. The component collects historical market prices, available demand indicators, weather observations, weather forecasts, and seasonal information. These data are aligned by crop, market, district, and date so that the model can learn how demand and price move over time. The initial crop scope includes high-priority fruits such as banana, mango, and papaya because these fruits are commercially important and sensitive to timing.

Data preprocessing is central to the methodology. Raw market records may contain missing days, inconsistent units, abnormal spikes, duplicate entries, or district naming differences. Weather observations may have different temporal resolution from market price reports. Seasonal information may be categorical rather than numerical. The preprocessing pipeline therefore fills missing values where justifiable, flags outliers, normalizes numerical variables, creates lag features, encodes day-of-week and seasonal effects, and constructs sliding time windows for sequence learning. This process converts heterogeneous data into a machine-learning-ready format.

The forecasting model is based on a multivariate LSTM design. LSTM is suitable because fruit demand and price can depend on longer temporal patterns rather than only on yesterday's price. For example, rainfall may affect harvesting, festivals may

affect buyer demand, and prior price momentum may influence farmer decisions. The model is compared conceptually against SARIMA and Prophet baselines to avoid assuming that a deep learning model is always superior. If a particular fruit has sparse data, a simpler baseline may be more stable. FreshRoute can therefore support model selection by crop, market, and data maturity.

The farmer-facing output is intentionally simple. The system should not show only numeric predictions to rural users. It converts forecasts into advisories such as SELL_SOON, DELAY_HARVEST, or NO_CHANGE, accompanied by trend direction, confidence, and a short explanation. These advisories are delivered through a mobile dashboard, Firebase push notification, and SMS fallback. Offline caching ensures that the most recent forecast remains visible even during intermittent connectivity. This design recognizes that the practical success of forecasting depends on interpretability and delivery, not only model accuracy.

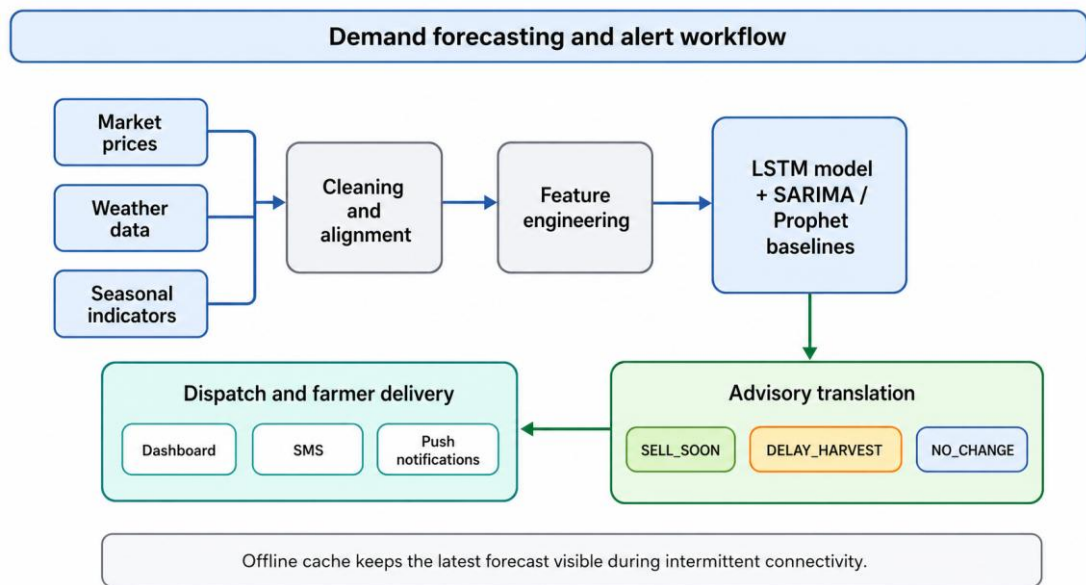


Figure 2.2: Demand forecasting and alert workflow.

Source: Prepared by the FreshRoute project team from the forecasting component methodology.

Table 2.5: Data sources and processing use.

Data category	Examples	Processing use
Market prices	Daily or weekly price records from economic centers and public reports.	Target variable, lag features, price trend, and volatility indicators.
Demand indicators	Buyer orders, historical sales, supply availability, and market arrivals where available.	Demand level estimation and buyer-interest context.
Weather observations	Rainfall, temperature, humidity, and weather alerts.	Features that influence harvest timing, transport risk, and demand shocks.

Seasonal indicators	Month, week, holidays, crop season, and production cycle.	Calendar features for repeated seasonal patterns.
User preferences	Crop, district, alert threshold, preferred channel, and language.	Personalized recommendations and notification routing.

Source: Prepared by the FreshRoute project team from the forecasting component report.

2.5 Blockchain marketplace and trust methodology

The marketplace component addresses the trust and bargaining problems that prevent direct farmer-buyer trade. A conventional digital listing service would not be sufficient because buyers need assurance of identity, quantity, quality, and reliability, while farmers need assurance of payment and fairness. FreshRoute therefore models the marketplace as a trust infrastructure rather than only a catalog. The proposed design uses verified onboarding, decentralized identity concepts, verifiable credentials, smart agreement lifecycle states, and aggregation logic.

Hyperledger Fabric was selected conceptually because it is a permissioned blockchain framework designed for known organizations and confidential business workflows [16]. In a prototype deployment, farmers, buyers, quality assessors, transport partners, and platform administrators can be represented by organizational identities and roles. Fabric channels and private data collections are suitable for B2B settings because not every participant should view every negotiated price or buyer procurement volume. CouchDB-style world state can support rich queries over current listings and agreements, while the ledger preserves immutable history.

The marketplace lifecycle begins with onboarding. After verification, a farmer receives a platform identity and a decentralized identifier record. When the farmer lists produce, the listing includes fruit type, quantity, grade, expected harvest date, location, and price expectations. A buyer places an order with quantity, grade, delivery date, destination, and maximum price. The matching and aggregation engine ranks available supply using location, quantity, grade, price, reliability, and reputation. Because smallholders may not individually satisfy institutional demand, the engine can combine several farmers into one fulfillment plan.

Once a buyer accepts a fulfillment plan, the system creates a Smart Sales Agreement. This agreement records the buyer order, selected farmer contributions, agreed price, delivery schedule, escrow or payment state, quality verification state, and settlement state. After delivery and quality confirmation, the system updates reputation and

creates verifiable evidence of performance. Reputation therefore becomes portable digital trust, replacing part of the informal trust currently held by intermediaries. Figure 2.3 shows the lifecycle at a high level.

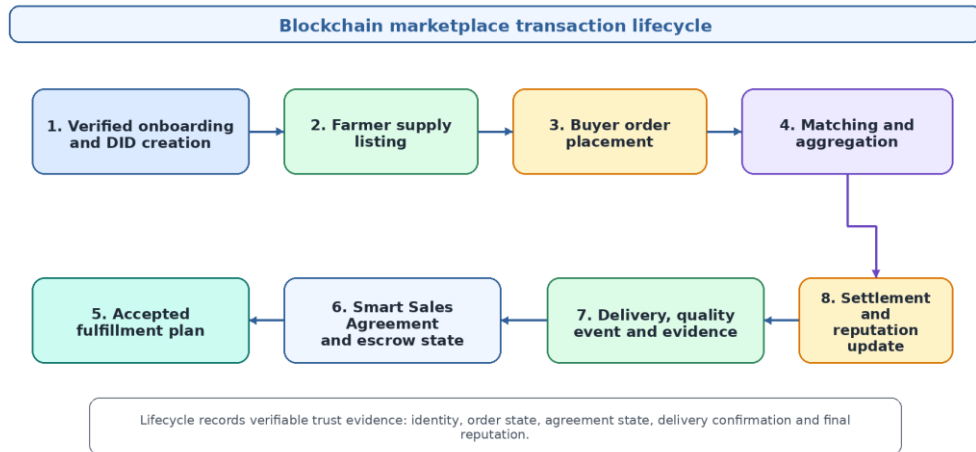


Figure 2.3: Blockchain marketplace transaction lifecycle.

Source: Prepared by the FreshRoute project team from the marketplace component design.

2.6 Spoilage-aware logistics and telemetry methodology

The logistics component addresses the physical vulnerability of fruit after trade commitments are made. Even if a forecast is accurate and a buyer order is fairly matched, losses can still occur if the fruit is moved in an unsuitable vehicle, delayed in extreme weather, mixed with incompatible loads, or left without condition monitoring. FreshRoute therefore treats logistics as a decision-support problem. The module receives matched orders, fruit attributes, pickup and destination locations, vehicle availability, capacity, vehicle type, and environmental conditions. It then recommends vehicle assignments and routes that reduce freshness risk.

Vehicle suitability is evaluated through explicit rules. Fruits may differ in fragility, thermal tolerance, ripeness stage, load size, and safe journey time. Vehicles may differ in capacity, availability, refrigeration capability, shift time, and location. The logistics engine therefore filters or ranks vehicles based on capacity fit, temperature sensitivity, distance, expected rain exposure, fruit attributes, and driver or vehicle availability. This makes the decision explainable: the system can state why a reefer is required, why an open vehicle is unsafe, or why a driver should not take another trip.

Routing is handled after suitability screening. The engine can use routing services such as OSRM to obtain travel distances and route estimates [28]. For prototype purposes, heuristic routing and local optimization are more practical than an exact large-scale vehicle routing solver. The key design goal is not only to minimize distance but to sequence pickup and delivery so that perishable loads remain within safe exposure windows. In the Matara district simulation described in the logistics component report, the engine reused refrigerated vehicles across multiple waves while observing shift constraints and maintaining feasibility under hot weather.

Telemetry provides operational visibility after a route is assigned. The module supports GPS, temperature, humidity, and status events from a vehicle or Arduino-style gateway. If a threshold is crossed, the alert service can notify the transporter, administrator, farmer, or buyer. Telemetry also creates evidence for traceability and complaint handling. If fruit quality deteriorates, stakeholders can examine whether unsafe temperature, delay, or route deviation occurred. Figure 2.4 summarizes the logistics workflow.

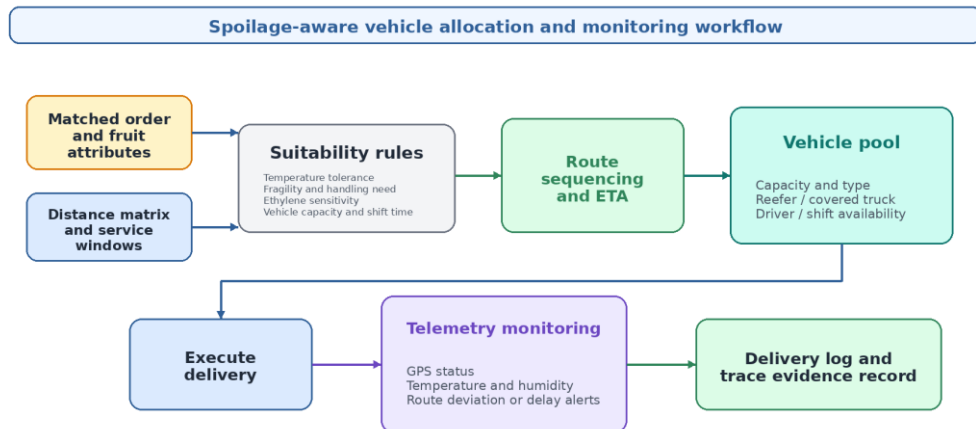


Figure 2.4: Spoilage-aware vehicle allocation and monitoring workflow.
 Source: Prepared by the FreshRoute project team from the logistics component design.

2.7 Quality grading and complaint validation methodology

The quality component addresses the subjectivity of manual grading and the lack of objective evidence in disputes. In conventional markets, fruit grade is often judged visually by humans. This can be fast and practical, but two people may grade the

same fruit differently. Buyers may reject goods after delivery, while farmers may argue that the fruit was damaged during transport. FreshRoute therefore includes an image-based grading component that can provide consistent visible-quality evidence.

The individual quality component uses mango as the case study fruit. Mango is suitable because important external quality cues are visible on the peel, including color stage, spots, blemishes, scratches, and surface marks. The initial grade-only classifier was improved into an explainable multi-attribute design. Instead of predicting only Grade A, Grade B, or Grade C, the model predicts color stage, spot severity, blemish severity, and final grade. This makes the grading decision easier to explain during evaluation and more useful for future complaint workflows.

The methodology begins with dataset redesign. Images are organized with a CSV label file that records image path, final grade, color stage, spot severity, blemish severity, and optional defect flags. Preprocessing includes background cleanup, crop refinement, resizing, augmentation, and normalization. A MobileNetV3-large backbone is used because it provides a practical balance between feature learning and deployment efficiency [14]. Four output heads predict the attributes and final grade. A rule-based grade mapping from attributes can be compared with the direct grade prediction. Grad-CAM can be used to visualize image regions that influenced the decision [15].

The component is intentionally limited to external visible quality from a single RGB image. It does not claim to measure sweetness, internal bruising, chemical residue, exact size, or hidden disease. This scope limitation is important because it prevents weak claims. Within its scope, the component provides a stronger and more explainable basis for quality verification than a black-box grade-only model.

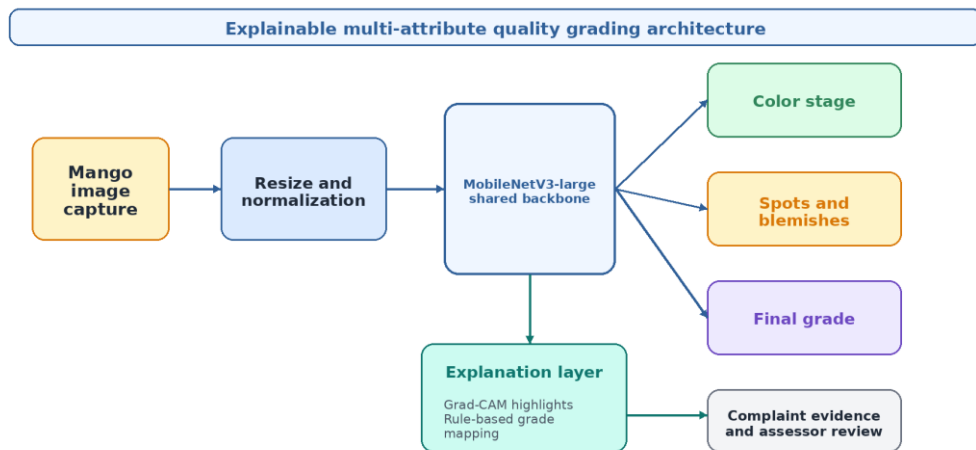


Figure 2.5: Explainable multi-attribute quality grading architecture.

Source: Prepared by the FreshRoute project team from the quality grading component design.

2.8 Algorithms, rules and integration logic

FreshRoute uses a combination of predictive models, deterministic business rules, and heuristic optimization. This hybrid approach is appropriate because not every part of the supply chain should be treated as a black-box machine-learning problem. Forecasting can learn temporal patterns, but payment settlement requires explicit state transitions. Logistics can use route estimates, but vehicle suitability must respect clear safety rules. Quality grading can use deep learning, but the final explanation benefits from explicit attribute labels and rule-based comparison.

The forecasting algorithm receives multivariate time-series windows and produces predicted demand or price values. These values are converted into advisory categories using threshold logic and confidence scoring. The matching algorithm ranks farmer listings against buyer orders using quantity, grade, location, price, timing, and reputation. The blockchain agreement logic moves through states such as created, accepted, escrowed, dispatched, delivered, verified, settled, or disputed. The logistics algorithm filters unsafe vehicles, ranks feasible candidates, sequences pickup and delivery, and listens for telemetry exceptions. The grading algorithm predicts visible attributes and compares direct grade output with rule-based grade mapping.

Integration logic ensures that module outputs become inputs for other modules. Forecasts can trigger farmer alerts and encourage early listing. Listings become marketplace supply. Accepted matches become logistics jobs. Logistics events become traceability evidence. Quality grades become marketplace attributes and complaint evidence. Completed transactions update reputation. Notifications connect all modules by alerting users when action is required. This design reduces silos and allows FreshRoute to behave as a system rather than as a collection of separate student components.

Table 2.6: Algorithms and decision rules.

Area	Method or rule	Purpose
Forecasting	Multivariate LSTM with SARIMA and Prophet baselines	Predict demand or price and compare model stability.
Advisory generation	Threshold and confidence rules	Translate numerical forecasts into simple farmer-facing recommendations.
Marketplace matching	Ranking and aggregation heuristic	Combine fragmented supply into buyer-ready fulfillment plans.
Smart agreement	State transition rules	Record order acceptance, escrow, delivery, verification, settlement, and dispute states.
Logistics allocation	Vehicle suitability filtering and ranking	Avoid unsafe vehicle choices for thermal, capacity, weather, or shift constraints.
Routing	Distance matrix and pickup-before-drop sequencing	Create executable routes for assigned vehicles.
Quality grading	MobileNetV3 multi-head classification and rule-based mapping	Predict visible quality attributes and compare explainable grade decisions.
Telemetry alerts	Threshold and event rules	Warn stakeholders about temperature, humidity, GPS, or delay exceptions.

Source: Prepared by the FreshRoute project team.

2.9 Tools, technologies and data management

The selected technology stack reflects both prototype practicality and integration needs. Node.js and Express are suitable for API orchestration because they support REST endpoints, middleware, external service integration, and rapid development. PostgreSQL or Supabase supports structured records such as users, listings, orders, vehicles, forecasts, and quality results [27]. Redis-style caching can reduce forecast retrieval latency. Hyperledger Fabric concepts support permissioned ledger logic and reputation evidence [16], [29]. FastAPI is suitable for serving machine-learning forecasts, while PyTorch and MobileNetV3 are suitable for image grading.

External services are used where building a full substitute would be inefficient. OSRM provides route and distance support [28]. Weather APIs provide environmental conditions that affect forecasting and logistics decisions. Firebase and SMS gateways such as Twilio provide notification channels [30], [32]. PayHere or a similar regulated gateway can support payment workflows in a production path [31].

This separation allows FreshRoute to focus on supply-chain logic while using established infrastructure for routing, communication, and payment.

Data management follows the principle that operational records should be structured, auditable, and role-protected. Farmer and buyer data should be stored with role-based permissions. Sensitive transaction details should be visible only to authorized parties. Quality images and supporting evidence should be stored with metadata and linked to the correct order or complaint. Telemetry events should preserve timestamps and vehicle identifiers. Forecast outputs should be versioned so that later evaluations can determine which prediction influenced a decision.

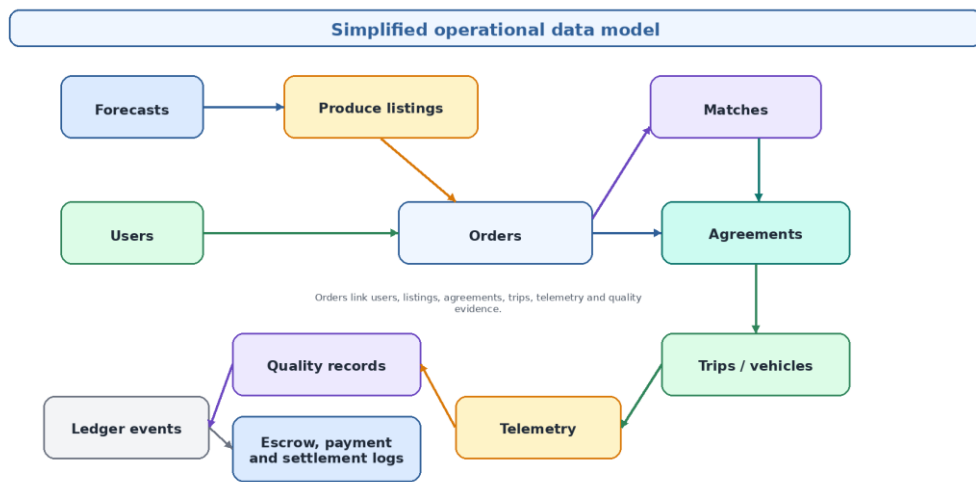


Figure 2.6: Simplified operational data model.

Source: Prepared by the FreshRoute project team.

Table 2.7: Tools and technologies.

Layer	Technology examples	Role in FreshRoute
Frontend	React, React Native, mobile dashboard concepts	Farmer, buyer, transporter, and admin user interfaces.
API layer	Node.js, Express, REST, HTTPS	Main service gateway and integration layer.
Forecasting	Python, FastAPI, LSTM, SARIMA, Prophet, Redis	Prediction service, baseline comparison, and fast forecast retrieval.
Data storage	PostgreSQL, Supabase, CouchDB world state	Operational records, user data, market data, and ledger world state.
Blockchain trust	Hyperledger Fabric, Fabric Gateway, DID/VC concepts	Transaction evidence, reputation, and smart agreement state.
Quality AI	PyTorch, MobileNetV3-large, ONNX, Grad-CAM	Image grading, attribute prediction, explanation, and model export.
Logistics	OSRM, weather APIs, GPS, Arduino gateway	Route calculation, environmental context, and telemetry.
Notifications/payment	Firebase, SMS gateway, PayHere	Alert delivery, payment authorization, and status communication.

Source: Prepared by the FreshRoute project team.

2.10 Commercialization aspect of the product

FreshRoute's commercialization potential is based on solving real pain points for both farmers and buyers. Farmers benefit from better market intelligence, direct access, transparent settlement, and quality-based trust. Buyers benefit from verified suppliers, traceable procurement, aggregated supply, and reduced counterparty risk. Transporters benefit from organized jobs and route guidance. Administrators and institutions benefit from anonymized supply-chain intelligence and structured evidence for dispute resolution.

The product can begin as a pilot-stage platform in one or two fruit-producing districts with a limited crop set. Early adoption should prioritize farmer organizations, cooperatives, institutional buyers, and transport partners who already face repeated sourcing problems. The initial revenue model should remain simple. Buyers or institutional users may pay a subscription or transaction service fee for reliable sourcing, traceability, and reports. Logistics providers may pay for job coordination once demand is stable. Farmer-facing access should be low cost or subsidized during early adoption because the platform's social value depends on including smallholders.

Commercialization must be gradual. The first stage is a minimum viable product with verified onboarding, listing, matching, logistics assignment, basic quality evidence, and notifications. The second stage is a district pilot with real transactions and measured KPIs. The third stage is partnership expansion with buyers, logistics firms, and agricultural agencies. The fourth stage is analytics and SaaS scaling for cooperatives and procurement teams. Throughout the process, FreshRoute should avoid overpromising; trust will depend on reliable operations, transparent fees, training, and fast support.

Table 2.8: Commercialization framework.

Element	Proposed approach	Rationale
Target users	Smallholder farmers, institutional buyers, transporters, cooperatives, administrators	Covers the actors required for end-to-end transaction execution.
Initial market	Selected districts and fruits such as banana, mango, and papaya	Limits operational complexity while testing high-value use cases.
Value proposition	Forecast-guided harvest, direct trade, verified reputation, freshness-aware delivery, quality evidence	Addresses waste, price opacity, trust, logistics, and complaints together.
Revenue model	Buyer subscription, transaction fee, premium analytics, logistics coordination fee	Charges users who receive procurement and coordination value while keeping farmer access affordable.
Rollout	MVP, district pilot, partner expansion,	Allows evidence-based growth and

	regional scale	operational learning.
Social impact	Improved farmer bargaining power, reduced waste, better transparency	Aligns commercialization with development goals.

Source: Prepared by the FreshRoute project team.

2.11 Testing and implementation

Testing is organized as a chain of evidence from low-level unit tests to integrated user workflows. The forecasting module should be evaluated using model accuracy metrics such as MAE, RMSE, MAPE, and WAPE, together with API response time and alert delivery success. The marketplace module should be evaluated through functional tests for onboarding, listing, ordering, matching, agreement creation, escrow state transitions, and reputation updates. The logistics module should be evaluated through rule tests, route assignment tests, synthetic benchmarks, and scenario simulations. The quality module should be evaluated through accuracy, precision, recall, F1-score, confusion matrices, and Grad-CAM evidence after final annotation.

Integration testing is critical because the value of FreshRoute depends on cross-module flow. A test scenario should begin with a forecast alert, continue with farmer listing, proceed to buyer order and matching, create a smart agreement, assign a vehicle, monitor delivery telemetry, record quality evidence, settle payment, and update reputation. If any module works alone but fails to share data, the integrated platform loses value. Therefore, tests must check not only individual functions but also shared identifiers, timestamps, status transitions, and error handling.

Implementation planning also includes deployment readiness. The prototype should be containerized or scripted so that services can be reproduced. Environment variables should hold secrets rather than hard-coded keys. Database migrations should be versioned. Model artifacts should be tagged with training metadata. API documentation should specify request and response formats. User acceptance testing should include farmer and buyer workflows with simple task descriptions. Figure 2.7 and Table 2.9 summarize the testing approach.

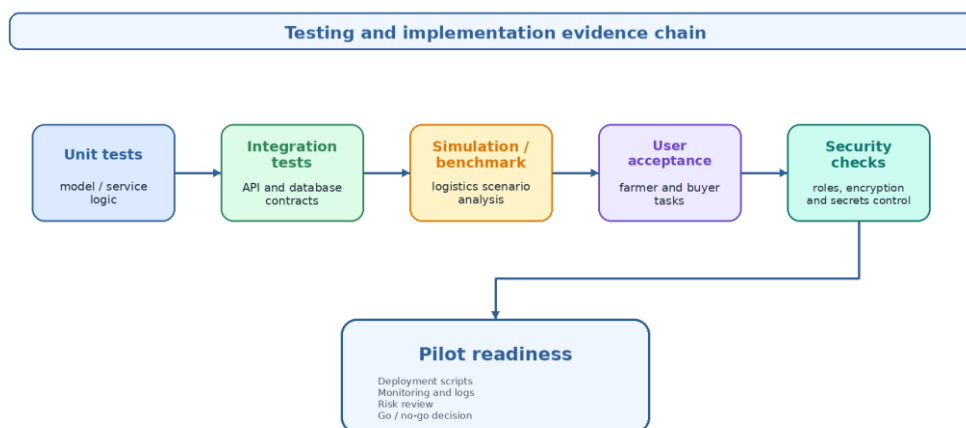


Figure 2.7: Testing and implementation evidence chain.

Source: Prepared by the FreshRoute project team.

Table 2.9: Testing and implementation strategy.

Test area	Example checks	Expected evidence
Forecasting	Data preprocessing, train/test split, LSTM versus baseline, forecast API, alert rule	Metric table, API logs, notification records, model report.
Marketplace	Onboarding, listing, order placement, matching, agreement, settlement, reputation	Functional test logs, ledger/state evidence, transaction lifecycle report.
Logistics	Vehicle suitability, route generation, telemetry threshold, benchmark, Matara-style scenario	Assignment outputs, runtime/spoilage/fulfillment metrics, telemetry alert log.
Quality grading	Annotation rubric, preprocessing, attribute heads, grade mapping, Grad-CAM	Classification reports, confusion matrices, sample explanations, export files.
Integration	End-to-end order-to-delivery flow	Workflow screenshot/log sequence and shared ID consistency.
Security	Role access, HTTPS/TLS, private data exposure, payment callback validation	Access-control tests, privacy review, security checklist.
Usability	Farmer listing, alert reading, buyer order, transporter update	Task completion rate, errors, qualitative feedback.

Source: Prepared by the FreshRoute project team.

2.12 Ethical, legal, security and operational considerations

FreshRoute processes data that can affect livelihoods and commercial relationships. Ethical design therefore requires informed consent during pilots, transparent communication about how data is used, and careful handling of personal and commercial information. Farmers should understand that forecasts are decision support rather than guarantees. Buyers should understand how reputation scores are created. Transporters should understand how telemetry is recorded. Administrators should avoid using platform data to unfairly exclude users without due process.

Legal and security issues are also significant. Payment settlement must comply with the requirements of regulated payment providers and local law. Personal data and commercial transactions must be protected from unauthorized access. Blockchain immutability must be balanced with privacy obligations; not all raw data should be

written permanently to a ledger. A practical design is to store only necessary proofs or hashes on-chain and retain sensitive files off-chain with appropriate access control. Role-based permissions, TLS, secure key storage, audit logs, and administrative review are necessary for production readiness.

Operational risks include limited internet connectivity, low digital literacy, insufficient vehicle supply, seasonal data gaps, model drift, user resistance, and dispute escalation. These risks do not make the project infeasible, but they require a pilot strategy that includes training, support, fallback workflows, manual override, and continuous improvement. Table 2.10 summarizes major risks and mitigations.

Table 2.10: Risk and mitigation summary.

Risk	Potential effect	Mitigation
Low connectivity	Farmers may miss alerts or fail to update listings.	Offline caching, SMS fallback, sync status, and simple retry logic.
Data quality gaps	Forecasts and quality models may perform poorly.	Source validation, outlier checks, model monitoring, and retraining plan.
Payment or trust failure	Users may return to intermediaries.	Clear settlement states, verified agreements, dispute workflow, and regulated payment integration.
Vehicle scarcity	Freshness-aware assignments may leave orders unfulfilled.	Fleet partnerships, overflow rules, priority ranking, and realistic delivery windows.
Security weakness	Sensitive trade data could be exposed.	Role-based access, TLS, secrets management, audit logs, and security testing.
User adoption barrier	Farmers or buyers may not trust the platform.	Training, pilot champions, transparent fees, local-language support, and gradual rollout.
Model overclaiming	Users may treat recommendations as certainty.	Confidence scores, explanations, disclaimers, and human override.

Source: Prepared by the FreshRoute project team.

2.13 Chapter summary

This chapter presented the methodology used to design FreshRoute as an integrated design-science artifact. It explained stakeholder requirements, architecture, forecasting, marketplace, logistics, quality grading, algorithms, tools, commercialization, testing, and ethical considerations. The methodology shows that FreshRoute is designed as a coordinated platform rather than a set of unrelated modules. The next chapter discusses the available prototype results, findings, limitations, and team contributions.

3 RESULTS AND DISCUSSION

3.1 Prototype implementation results

The FreshRoute prototype demonstrates a multi-module design in which forecasting, marketplace, logistics, and quality services are intended to cooperate through shared records and operational events. The individual thesis artifacts show that each member focused on a distinct component while preserving the common project vision. The forecasting component produced the data and alert design for farmer decision support. The marketplace component produced the direct trade, identity, reputation, and smart agreement design. The logistics component produced the spoilage-aware assignment, routing, and telemetry design. The quality component produced the explainable grading design for mango images.

The strongest evidence of implementation maturity is visible in the logistics and integrated backend review. The logistics component inspected a Node.js/Express backend in which route groups, controllers, services, schedulers, and external service integrations were visible. It documented links among forecasting services, matching logic, logistics engine functions, quality services, payment flows, notification routes, and traceability concepts. This indicates that FreshRoute moved beyond a paper-only design into a modular prototype. However, maturity varies by module. Forecasting and quality components define clear evaluation and output structures, while final field metrics and fully frozen model results still need to be inserted after final training and pilot validation.

The implementation should therefore be interpreted as a credible proof-of-concept rather than a production system. The project establishes the technical structure, decision logic, data flows, and evaluation plan required for a real pilot. It also reveals important deployment gaps: security hardening, reproducible packaging, complete user-interface testing, finalized model metrics, field data collection, and regulated payment integration. Table 3.1 summarizes implementation evidence by module.

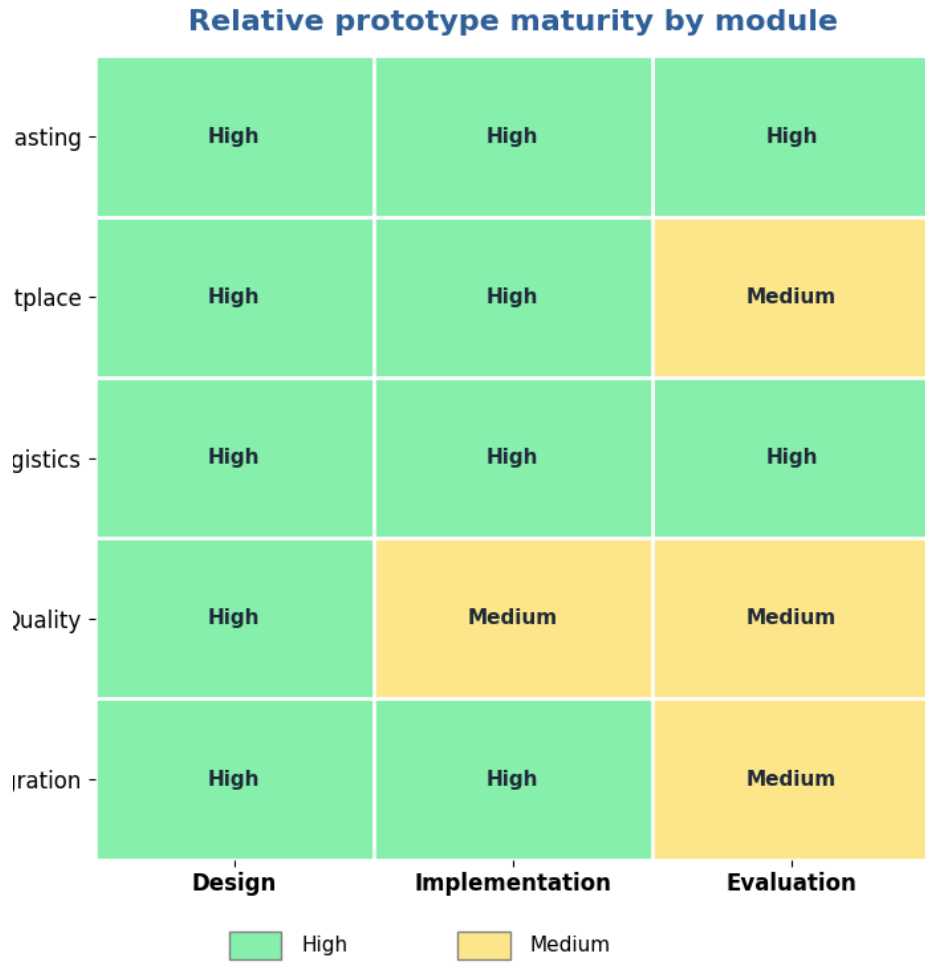


Figure 3.1: Relative prototype maturity by module.

Source: Prepared by the FreshRoute project team from component status evidence.

Table 3.1: Prototype implementation evidence by module.

Module	Available evidence	Interpretation
Demand forecasting	Data pipeline design, LSTM methodology, baseline comparison plan, FastAPI/PostgreSQL/Redis architecture, SMS/FCM alert workflow.	Methodology and evaluation design are clear; final numeric model metrics depend on final dataset execution.
Marketplace	Hyperledger Fabric architecture, DID/VC reputation concept, matching/aggregation logic, smart agreement lifecycle, functional and KPI test plan.	Strong trust framework; production settlement requires compliance and live pilot validation.
Logistics	Vehicle suitability rules, route generation, telemetry gateway, synthetic benchmark, Matara district simulation.	Most concrete execution evidence; benchmark indicates safer allocation than simple baselines.
Quality grading	CSV annotation rubric, MobileNetV3 multi-head architecture, direct/rule grade logic, Grad-CAM output plan.	Explainability design solves grade-only weakness; final performance requires completed annotation and training run.
Integration	Shared platform architecture and service-oriented backend concepts.	System design is coherent; end-to-end pilot evidence remains future work.

Source: Prepared by the FreshRoute project team from individual reports [34], [35].

3.2 Module-level evaluation results

The forecasting component is evaluated through data preparation quality, model accuracy, baseline comparison, API performance, alert reliability, offline usability, and user acceptance. The draft forecasting results correctly avoid inventing final field values before final model testing. Instead, it defines the metrics that should be reported: MAE, RMSE, MAPE, WAPE, API response time, alert delivery success rate, and task completion rate. This is appropriate because an agricultural forecasting system must be judged both by prediction error and by whether the prediction reaches farmers in an understandable form.

The marketplace component is evaluated through prototype workflow completion, transaction performance, usability feedback, and economic impact estimates. The expected evidence includes onboarding, DID creation, supply listing, buyer order placement, matching, smart agreement creation, delivery confirmation, settlement, and reputation update. For performance, read queries and write transaction commitment times should be measured under increasing concurrency. For economic impact, the platform should estimate farmer share improvement, payment delay reduction, and transparency gains, while clearly distinguishing simulated estimates from pilot evidence.

The logistics component provides the most concrete numerical evidence available in the submitted artifacts. A synthetic benchmark generated 10,000 mock orders, a heterogeneous fleet of 2,500 vehicles, and three assignment strategies: random selection, naive-greedy selection, and FreshRoute allocation. Across five runs, FreshRoute recorded an average spoilage rate of 0.00 percent under the benchmark's rule-based definition, compared with 13.29 percent for random assignment and 12.78 percent for naive-greedy assignment. FreshRoute also achieved a slightly higher average fulfillment rate of 21.47 percent compared with 20.58 percent and 20.67 percent respectively. The trade-off was runtime: FreshRoute averaged approximately 733 milliseconds compared with 270 milliseconds for random assignment and 256 milliseconds for naive-greedy assignment.

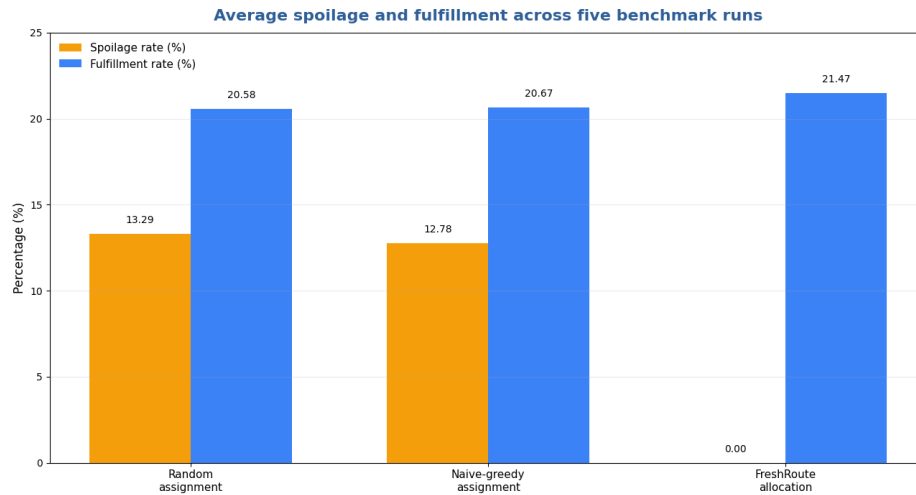
These logistics results should be interpreted carefully. The benchmark does not measure real biochemical fruit spoilage; it measures unsafe assignment conditions according to the prototype's rules. Nevertheless, the comparison is meaningful because all algorithms operated on the same generated workloads. The result indicates that explicit perishability rules add decision quality beyond naive assignment. At the same time, fulfillment remains low because the synthetic workload and fleet constraints create many infeasible orders. A field pilot would be required to determine actual loss reduction, cost impact, and transporter adoption.

The quality grading component evaluates a redesigned task definition rather than presenting final numerical performance. The major result is the move from a simple three-folder Grade A/B/C classifier to a richer CSV annotation structure that records color stage, spot severity, blemish severity, final grade, and optional defect flags. This enables direct grade prediction and rule-based grade comparison. Evaluation metrics should include accuracy, precision, recall, F1-score, confusion matrix analysis, and Grad-CAM review. This design makes the grading component more defensible and useful for future complaint validation.

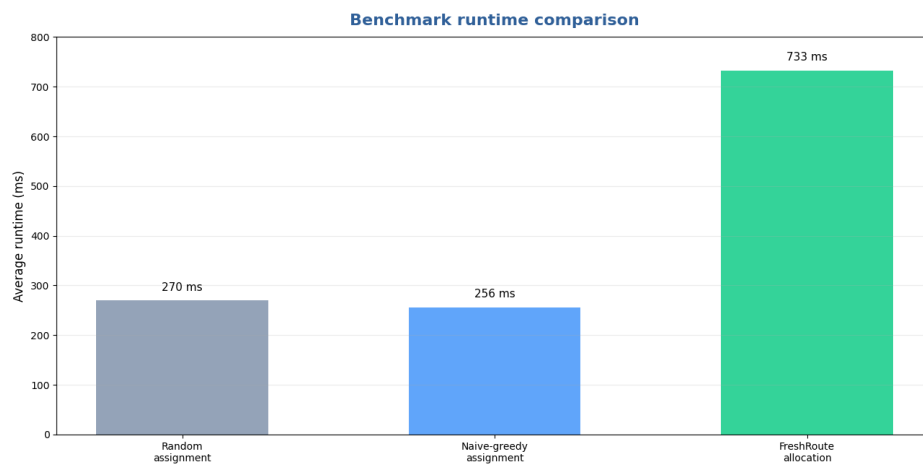
Table 3.2: Average benchmark results over five logistics runs.

Algorithm	Average spoilage rate (%)	Average fulfillment rate (%)	Average runtime (ms)	Interpretation
Random assignment	13.29	20.58	270	Fast but unsafe because vehicles are not selected using fruit-specific rules.
Naive-greedy assignment	12.78	20.67	256	Fast and slightly more structured, but still violates perishability constraints.
FreshRoute allocation	0.00	21.47	733	Eliminates unsafe allocations in the benchmark while improving fulfillment slightly at higher runtime.

Source: Averaged from the logistics component benchmark results [35].



*Figure 3.2: Average spoilage and fulfillment across five benchmark runs.
Source: Prepared by the FreshRoute project team from logistics benchmark averages.*



*Figure 3.3: Benchmark runtime comparison.
Source: Prepared by the FreshRoute project team from logistics benchmark averages.*

Table 3.3: Integrated evaluation findings.

Evaluation area	Finding	Implication
Forecasting	The evaluation framework considers accuracy, API performance, alert reliability, and usability.	A useful forecast service must be judged by delivery and understanding, not only by error metrics.
Marketplace	Trust, reputation, aggregation, and settlement are central to adoption.	A listing-only marketplace would not solve the intermediary dependency problem.
Logistics	FreshRoute allocation removed unsafe assignments in the benchmark rules.	Perishability-aware rules improve decision quality but require field validation.
Quality grading	Attribute-based labels improve explanation compared with grade-only classification.	Quality decisions become easier to defend and integrate with complaints.
System integration	Module outputs are designed to become downstream inputs.	Integrated architecture is the project's main group contribution.

Source: Prepared by the FreshRoute project team.

3.3 Integrated platform findings

The first integrated finding is that post-harvest loss cannot be addressed effectively through a single isolated feature. A forecasting model may help farmers plan harvests, but it will not prevent loss if orders, vehicles, quality, and payments remain informal. A blockchain marketplace may support direct trade, but it cannot preserve fruit if logistics is unmanaged. A quality model may improve grading, but it cannot establish responsibility without transport and transaction evidence. FreshRoute's system-level contribution is therefore the coordination of multiple decision points across the chain.

The second finding is that trust is a data and workflow problem. In the traditional chain, trust is often held by intermediaries because they know farmers, buyers, transporters, and local market norms. FreshRoute attempts to digitize this trust by recording identity, reputation, quality evidence, delivery status, and settlement history. This does not mean that human relationships disappear. Rather, the platform creates verifiable evidence so that farmers and buyers can expand direct trading relationships beyond immediate personal networks.

The third finding is that explainability matters across all modules. Farmers need to know why an alert recommends selling soon. Buyers need to know why a farmer is ranked highly. Transporters need to know why a refrigerated vehicle is required. Administrators need to know why a quality dispute is accepted or rejected. The platform therefore avoids relying purely on opaque scores. Forecast confidence, matching criteria, suitability rules, attribute-based grading, and trace records all support explainability.

The fourth finding is that prototype readiness is not the same as deployment readiness. FreshRoute has a coherent architecture and strong component designs, but production deployment requires additional work. Final datasets must be expanded and cleaned. User interfaces must be tested with low-literacy users. Payment flows must be compliant. Blockchain privacy design must be audited. Logistics must be validated with real vehicles and sensors. The project is therefore best positioned as a

pilot-ready research prototype after further hardening, not as an immediately deployable national platform.

Table 3.4: Research question findings matrix.

Research question	Finding
RQ1: Integrated platform support	The system architecture shows that forecasting, marketplace, logistics, and quality modules can be linked through shared records and status events.
RQ2: Demand forecasting	The forecasting methodology defines a practical pipeline from market/weather data to LSTM predictions and farmer-facing alerts.
RQ3: Marketplace trust	A direct marketplace needs reputation, verified identity, aggregation, and smart agreement states to substitute for intermediary trust functions.
RQ4: Logistics freshness	Benchmark evidence indicates that explicit vehicle suitability rules can eliminate unsafe allocations under tested conditions.
RQ5: Quality explanation	Multi-attribute grading makes quality decisions more interpretable than grade-only classification.
RQ6: Prototype readiness	The platform is a credible proof-of-concept but requires field pilots, security review, and final model validation.

Source: Prepared by the FreshRoute project team.

3.4 Discussion

The major strength of FreshRoute is architectural integration. Many agricultural applications solve only one part of the chain: price information, listing, weather advisory, or transport booking. FreshRoute combines these into a closed-loop workflow. This matters because fruit loss and farmer disadvantage occur across the whole sequence of decisions. The group project therefore demonstrates a strong understanding of the problem environment and a practical software engineering response.

A second strength is the balance between intelligent models and rule-based control. The forecasting and grading components use machine learning where prediction from data is valuable. The logistics and marketplace components use explicit rules where accountability, explainability, and safety are required. This hybrid design is suitable for a rural supply-chain context because stakeholders need recommendations they can understand and dispute if necessary.

A third strength is attention to rural accessibility. The forecasting component includes SMS fallback and offline concepts. The marketplace component considers direct trade for smallholders rather than only large farms. The logistics component recognizes limited refrigerated vehicles and hot-weather constraints. The quality

component limits itself to visible external grading rather than claiming hidden internal quality. These design choices indicate sensitivity to context.

The limitations are equally important. Forecasting performance depends on data volume, quality, and representativeness. Marketplace adoption depends on whether farmers and buyers trust the platform enough to change behavior. Blockchain adds credibility but also complexity, key-management issues, and governance questions. Logistics benchmarks simplify real-world deterioration, road conditions, fuel costs, and driver behavior. Quality grading depends on annotation consistency, lighting, camera quality, and dataset diversity. Security and privacy controls need deeper testing before production.

There are also validity threats associated with using prototype artifacts. Some evidence comes from design documents and planned evaluation structures rather than final field measurements. Some implementation details may be snapshots rather than final production code. The logistics benchmark uses generated data. The quality and forecasting components avoid fabricating unmeasured results, which is academically responsible but leaves some evaluation sections incomplete until the final run. These limitations should guide future work rather than weaken the central contribution.

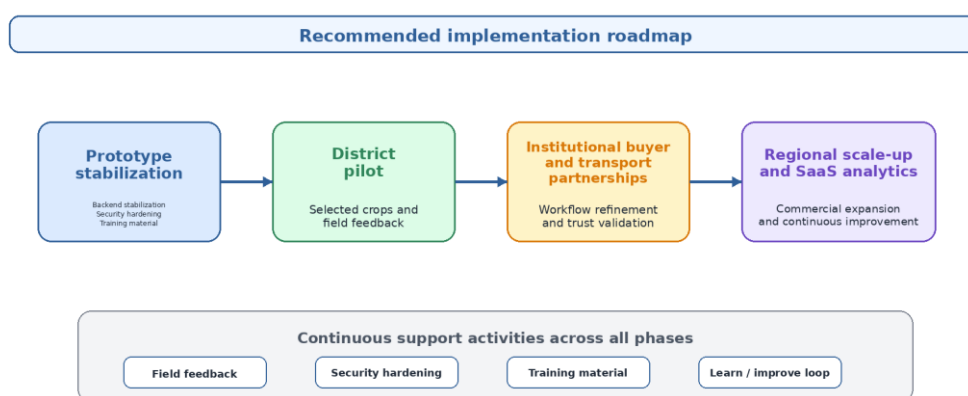


Figure 3.4: Recommended implementation roadmap.

Source: Prepared by the FreshRoute project team.

3.5 Summary of each student's contribution

FreshRoute was developed as a group project with clearly differentiated individual contributions. Aashani Dulanjana Samarakoon contributed the AI-driven demand

forecasting and alert component. This work defined how market, weather, and seasonal data can be converted into LSTM-based forecasts and farmer-facing advisories delivered through mobile and SMS channels.

Dias N. A. contributed the blockchain-powered direct marketplace and trust component. This work defined the direct farmer-buyer trading model, Hyperledger Fabric architecture, DID/VC-based reputation, matching and aggregation logic, and smart agreement lifecycle for trusted settlement and traceability.

Gallearachchi S. C. contributed the spoilage-aware logistics and telemetry component. This work defined vehicle suitability rules, route generation, telemetry monitoring, alerts, and benchmark/simulation analysis for reducing unsafe transport assignments and improving delivery visibility.

Sudarshan Raveendranath Dayarathna contributed the explainable fruit quality grading component. This work redesigned the mango grading dataset, defined visible attribute labels, selected MobileNetV3-large, designed the multi-head model, and planned Grad-CAM explanation outputs for future quality verification and complaint validation.

Table 3.5: Summary of each student's contribution.

Student	Student ID	Primary component	Key contribution
Aashani Dulanjana Samarakoon	IT22124494	Demand forecasting and alerts	LSTM forecasting methodology, data preprocessing, API design, offline-friendly dashboard, SMS and push alerts.
Dias N. A.	IT22213730	Blockchain marketplace and trust	Direct marketplace design, Hyperledger Fabric concept, DID/VC reputation, matching, aggregation, and smart agreement lifecycle.
Gallearachchi S. C.	IT22230874	Spoilage-aware logistics	Vehicle suitability rules, route optimization workflow, telemetry alerts, logistics benchmark, and Matara scenario analysis.
Sudarshan Raveendranath Dayarathna	IT22257154	Quality grading and complaint support	Mango grading rubric, MobileNetV3 multi-head architecture, attribute labels, rule-based grade mapping, and Grad-CAM plan.

Source: Prepared by the FreshRoute project team from the individual component thesis documents.

3.6 Chapter summary

This chapter presented the available results and discussion for FreshRoute. It showed that the platform has a coherent multi-module prototype design, with the strongest

numerical evidence currently available from the logistics benchmark. It also discussed how forecasting, marketplace, logistics, and quality components support one another, while identifying limitations that must be addressed before production deployment. The next chapter concludes the thesis and provides recommendations for future work.

4 CONCLUSION AND RECOMMENDATIONS

4.1 Conclusion

This group thesis presented FreshRoute, a smart fruit supply chain system designed to reduce waste and eliminate excessive dependency on opaque middlemen in Sri Lanka. The research began from the observation that fruit losses, price opacity, delayed payments, weak logistics coordination, and subjective quality disputes are not independent problems. They occur in the same chain and reinforce one another. FreshRoute responds by connecting four components into one coordinated platform: AI demand forecasting, blockchain-based direct marketplace trust, spoilage-aware logistics, and explainable fruit quality grading.

The study shows that an integrated design is feasible at prototype level. The forecasting component provides a method for turning market, weather, and seasonal data into farmer-facing harvest advisories. The marketplace component provides a structure for direct farmer-buyer trade using verified identity, reputation, matching, aggregation, and smart agreement states. The logistics component provides freshness-aware vehicle allocation, routing, and telemetry. The quality component provides a more explainable approach to image-based grading by using visible attributes rather than only a final grade label.

The available evaluation evidence supports the value of the approach while also showing the need for further validation. The logistics benchmark indicates that FreshRoute's allocation rules can remove unsafe vehicle assignments under synthetic benchmark conditions and slightly improve fulfillment relative to random and naive baselines. Other components provide strong methodology and evaluation structures, but final field metrics, final model results, and live user evidence must be completed before production claims are made. The overall conclusion is that FreshRoute is a credible and context-sensitive proof-of-concept for a Sri Lankan smart fruit supply chain platform.

4.2 Recommendations

The first recommendation is to conduct a controlled district pilot with a small set of crops, farmers, buyers, and transport partners. The pilot should measure actual

harvest decisions, listing outcomes, transaction completion, payment time, delivery conditions, quality disputes, user satisfaction, and spoilage proxies. This will convert the current prototype evidence into field evidence.

The second recommendation is to finalize data pipelines and model evaluation. Forecasting requires cleaned historical data, reproducible train-test splits, baseline comparisons, and monitoring for model drift. Quality grading requires final annotation review, dataset balancing, model training, confusion matrix analysis, and Grad-CAM review by domain stakeholders. Without strong data governance, advanced models may produce unreliable recommendations.

The third recommendation is to harden the platform technically. Authentication, role-based access, TLS, secrets management, audit logs, payment callback validation, blockchain privacy design, and backup procedures should be reviewed before any real commercial deployment. The system should also include clear error handling, retry logic, and manual override for rural operational conditions.

The fourth recommendation is to design adoption support around people, not only software. Farmers and transporters may need training, local-language interfaces, simple icons, SMS support, and support channels. Buyers may need procurement reports, traceability exports, and integration with existing purchasing processes. Platform trust will grow only if users experience practical value and fair governance.

4.3 Future work

Future work should expand FreshRoute from prototype to pilot-ready platform. The forecasting component can add more crops, market centers, weather variables, confidence explanations, and hybrid model selection. The marketplace component can improve reputation scoring, privacy-preserving credential exchange, buyer procurement dashboards, and regulated payment integration. The logistics component can incorporate real-time traffic, live weather feeds, dynamic vehicle availability, richer driver constraints, and field sensor validation. The quality component can expand beyond mango, support multiple image angles, include quality assessor review, and integrate before-versus-after complaint comparison.

The platform should also explore institutional partnerships. Farmer organizations, cooperatives, agricultural agencies, economic centers, and buyers could provide data, training, and adoption pathways. A phased commercialization model should begin with one region and limited crops before scaling. In the long term, FreshRoute can become a broader agricultural intelligence and coordination platform if it maintains the principles established in this thesis: transparency, explainability, rural accessibility, operational realism, and evidence-based improvement.

Table 4.1: Key conclusions and recommendations.

Area	Conclusion	Recommendation
Forecasting	Forecast-guided advisories can reduce information asymmetry before harvest.	Complete final model runs and validate with farmers.
Marketplace	Direct trade requires trust infrastructure, not only listings.	Pilot verified onboarding, reputation, aggregation, and settlement workflows.
Logistics	Perishability-aware assignment improves safety under benchmark conditions.	Validate with real vehicles, routes, sensors, and delivery outcomes.
Quality	Attribute-based grading improves explainability.	Complete annotation, training, and stakeholder review of quality outputs.
Integration	FreshRoute's main contribution is closed-loop coordination.	Stabilize APIs, shared data models, security, and end-to-end tests before pilot.

Source: Prepared by the FreshRoute project team.

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GLOSSARY

Table G.1: Glossary of key terms.

Term	Meaning in this thesis
Aggregation	Combining produce from multiple smallholder farmers into a buyer-ready order volume.
Attribute head	A model output branch that predicts a visible fruit-quality attribute such as color stage or spot severity.
Cold chain	A temperature-controlled handling and transport process used to preserve perishable products.
Decentralized Identifier	A digital identity concept where the subject controls a verifiable identifier rather than relying only on a centralized account.
Design Science Research	A research approach that builds and evaluates an artifact to solve a real problem.
Forecast advisory	A farmer-facing recommendation derived from a model prediction, such as SELL SOON or DELAY HARVEST.
Grad-CAM	A visualization technique that highlights image regions influencing a neural network prediction.
Intermediary	An actor such as a collector or trader who connects farmers to markets and may control information or negotiation.
Post-harvest loss	Quantity or quality loss that occurs after harvest and before final consumption.
Reefer vehicle	A refrigerated vehicle used for cold-chain transportation.
Smart Sales Agreement	A digital agreement state flow that records order acceptance, delivery, quality verification, and settlement.
Telemetry	Remote measurement of conditions such as GPS location, temperature, and humidity during transport.
Verifiable Credential	A tamper-evident digital claim about identity, delivery, quality, or reputation.

Source: Prepared by the FreshRoute project team.

The glossary is included to support readers from both technical and non-technical backgrounds. Several terms in FreshRoute combine agriculture, software engineering, logistics, and blockchain concepts, and short definitions improve readability.

APPENDICES

Appendix A: Functional requirement matrix

Table A.1: Detailed functional requirement matrix.

ID	Requirement	Input / trigger	Expected output
FR1	Forecast retrieval	Farmer requests forecast by crop, district, market, and date range.	Forecast service returns trend, confidence, and advisory.
FR2	Alert threshold	Farmer configures price or demand threshold.	Notification is created when forecast crosses threshold.
FR3	Farmer registration	Farmer submits identity and farm details.	Administrator can approve account and role.
FR4	Produce listing	Farmer lists fruit type, quantity, grade, price, harvest date, and location.	Listing appears for matching when active.
FR5	Buyer order	Buyer submits crop, quantity, grade, price ceiling, date, and destination.	Order is stored and ready for matching.
FR6	Matching	System ranks and aggregates eligible farmer listings.	Buyer receives a fulfillment proposal.
FR7	Agreement	Buyer and farmer accept terms.	Smart agreement state is created and traceable.
FR8	Vehicle allocation	Order requires transport.	System assigns feasible vehicle or flags shortage.
FR9	Telemetry	Vehicle transmits GPS and condition data.	Alerts are raised for threshold violations.
FR10	Quality grading	Fruit image is submitted for grading.	System predicts attributes and grade with explanation evidence.
FR11	Payment status	Payment authorization or settlement is updated.	Stakeholders can view payment state.
FR12	Reputation update	Delivery and quality are verified.	Farmer reputation evidence is updated.

Source: Prepared by the FreshRoute project team.

Appendix B: API and service inventory

Table B.1: API and service inventory.

Service group	Representative endpoints or actions	Purpose
Authentication	register, login, role verification, profile update	User access control for all modules.
Forecasting	get forecast, archive prices, retrieve advisory, alert configuration	Farmer decision support and forecast storage.
Marketplace	create listing, update stock, create order, generate match, confirm proposal	Direct trade and aggregation workflow.
Agreement	create agreement, update escrow, delivery confirmation, settlement status	Stateful evidence for transaction execution.
Logistics	vehicle registry, assign vehicle, route request, delivery status, telemetry ingest	Freshness-aware transport coordination.
Quality	image upload, grade prediction, complaint evidence, assessor review	Quality verification and dispute support.
Notification	SMS dispatch, push notification, alert log, user preferences	Multi-channel communication.
Administration	user approval, dispute management, report dashboard, configuration	Governance and monitoring.

Source: Prepared by the FreshRoute project team.

Appendix C: Test case matrix

Table C.1: Test case matrix.

Test ID	Scenario	Input condition	Expected result
TC1	Forecast API success	Valid crop and district	Forecast response includes trend, value, confidence, and advisory.
TC2	Forecast invalid input	Unknown crop or missing district	API returns validation error without server failure.
TC3	SMS alert rule	Forecast crosses configured threshold	Notification log created and SMS dispatch attempted.
TC4	Farmer listing	Authenticated farmer submits valid produce listing	Listing stored and visible to matching service.

TC5	Buyer order	Authenticated buyer submits valid demand request	Order stored with pending match state.
TC6	Matching aggregation	Multiple farmer listings partially satisfy order	System generates ranked fulfillment plan.
TC7	Agreement state	Buyer confirms fulfillment proposal	Agreement moves to accepted or escrow-pending state.
TC8	Vehicle allocation	Perishable fruit requires refrigerated transport	System selects reefer or flags no safe vehicle.
TC9	Telemetry alert	Temperature exceeds safe limit	Alert is created for transporter/admin and stored in event log.
TC10	Quality grading	Valid mango image submitted	Model returns color, spot, blemish, and grade outputs.
TC11	Complaint review	Buyer submits complaint with delivery and image evidence	Administrator can inspect quality, telemetry, and transaction records.
TC12	Role access	Farmer attempts admin-only action	Request is denied with appropriate authorization error.

Source: Prepared by the FreshRoute project team.

Appendix D: Indicative pilot budget

Table D.1: Indicative pilot-stage budget.

Budget item	Estimated amount (LKR)	Purpose
Cloud hosting and database	60,000	Prototype hosting, database, storage, and backups for pilot period.
SMS and notification costs	30,000	SMS fallback for farmers and alert testing.
Field training and materials	75,000	Training sessions, printed guides, and local-language material.
Transport sensor kits	120,000	GPS/temperature/humidity test devices for selected vehicles.
Travel and stakeholder meetings	80,000	District visits, buyer meetings, and farmer organization coordination.
Quality image collection	40,000	Image capture sessions and annotation review.
Security and deployment review	50,000	Basic security testing, environment hardening, and documentation.
Contingency	45,000	Unexpected field or technical costs.

Source: Prepared by the FreshRoute project team for planning purposes.

Appendix E: Risk register

Table E.1: Risk register and mitigation plan.

ID	Risk	Likelihood	Impact	Mitigation
R1	Inaccurate forecast due to sparse data	Medium	High	Use baselines, confidence scores, model monitoring, and data expansion.
R2	Farmer resistance to digital workflows	Medium	High	Provide training, SMS fallback, and cooperative-level onboarding.
R3	Buyer reluctance to direct sourcing	Medium	High	Emphasize verified reputation, aggregation, traceability, and settlement status.
R4	Insufficient refrigerated vehicles	High	Medium	Prioritize sensitive loads, partner with logistics providers, and flag infeasible orders early.
R5	Quality model bias from limited images	Medium	Medium	Improve dataset diversity, annotation review, and validation by grade.
R6	Payment compliance complexity	Medium	High	Use regulated gateway integration and legal review before production settlement.
R7	Data privacy failure	Low	High	Apply role-based access, encryption, audit logs, and data minimization.
R8	Over-dependence on external APIs	Medium	Medium	Cache essential data, implement fallback paths, and monitor service availability.

Source: Prepared by the FreshRoute project team.

Appendix F: Pilot data collection plan

Table F.1: Pilot data collection plan.

Data area	Variables	Collection method	Timing
Farmer profile	Crop, district, farm size, phone type, current selling channel	Registration form and interview	Pilot start
Market decision	Harvest date, listing date, advisory received, action taken	Application log and farmer follow-up	Weekly
Transaction	Order quantity, price, buyer, agreement state, payment state	Platform database	Per transaction
Logistics	Vehicle type, route, pickup time, drop time, temperature, humidity, alerts	Telemetry and transporter app	Per trip
Quality	Initial grade, delivery grade, complaint status, image evidence	Quality service and admin review	Per batch
Usability	Task completion, error count, satisfaction, comments	UAT session and questionnaire	Pilot midpoint and end
Impact	Payment delay, unsold quantity, spoilage proxy, farmer perceived price fairness	Survey and transaction analysis	Pilot end

Source: Prepared by the FreshRoute project team.

Appendix G: Work breakdown and timeline

This appendix converts the group thesis scope into an implementation work breakdown. The sequence is designed for a pilot-ready version of FreshRoute after the academic prototype. The timeline should be adjusted after confirming partner availability, data readiness, and payment gateway compliance.

Table G.2: Work breakdown and timeline.

No.	Work package	Main activity	Indicative timing	Responsibility
1	Project initiation	Confirm pilot district, crops, partners, and success metrics.	Week 1	All members
2	Data consolidation	Clean market, weather, user, route, and image datasets.	Weeks 1-3	Forecasting and quality leads
3	Backend stabilization	Freeze API routes, database schema, logging, and configuration.	Weeks 2-4	Marketplace and logistics leads
4	Forecast model run	Train baseline and LSTM models and create forecast reports.	Weeks 3-5	Forecasting lead
5	Quality model run	Complete annotation review and train multi-head grading model.	Weeks 3-5	Quality lead
6	Marketplace workflow	Finalize onboarding, listing, order, matching, agreement, and reputation states.	Weeks 4-6	Marketplace lead
7	Logistics workflow	Finalize vehicle registry, assignment, telemetry ingest, and alert rules.	Weeks 4-6	Logistics lead
8	Integration testing	Execute end-to-end forecast-to-settlement pilot scenarios.	Weeks 6-7	All members
9	Security review	Check roles, secrets, TLS, privacy exposure, and payment callbacks.	Week 7	All members with supervisor review
10	UAT and training	Run user tasks with farmers, buyers, transporters, and admin users.	Weeks 8-9	All members
11	Pilot launch	Run limited real or supervised transactions and collect metrics.	Weeks 10-12	All members and partners
12	Evaluation report	Analyze metrics, update thesis evidence, and prepare improvements.	Week 13	All members

Source: Prepared by the FreshRoute project team.

Appendix H: Sample user workflows

The following workflow scenarios can be used during user acceptance testing. Each scenario should be executed with realistic data and observed by a facilitator who records completion time, errors, clarifying questions, and user comments.

Table H.1: Sample user workflow scenarios.

Workflow	Task sequence	Success criterion
Farmer forecast workflow	Open dashboard, select crop and district, view trend and advisory, decide whether to list produce.	User can interpret the advisory and explain the intended action.
Farmer listing workflow	Create a listing with fruit type, quantity, grade, price expectation, harvest date, and pickup location.	Listing is submitted without admin assistance and appears as available supply.
Buyer order workflow	Create buyer order, review matched farmer group, inspect reputation evidence, and confirm agreement.	Buyer understands the fulfillment plan and settlement state.
Transporter workflow	View assigned job, accept route, update pickup status, transmit telemetry, and complete delivery.	Transporter follows job sequence and understands alert messages.
Quality workflow	Capture or upload fruit image, obtain attribute and grade output, and review explanation evidence.	User understands why the fruit received the grade.
Admin dispute workflow	Open complaint, inspect order, quality, telemetry, and settlement records, and record resolution.	Administrator can justify decision using evidence rather than opinion only.

Source: Prepared by the FreshRoute project team.

Appendix I: Data dictionary

Table I.1: Data dictionary for core entities.

Field	Type	Description	Module
user_id	UUID/String	Unique identifier for platform user.	Users
role	Enum	Farmer, buyer, transporter, administrator, or quality assessor.	Users
crop_type	String	Fruit category such as banana, mango, or papaya.	Listings, forecasts, orders
quantity_kg	Decimal	Available or requested quantity in kilograms.	Listings, orders, trips
grade	Enum	Grade A, Grade B, Grade C, or quality attribute result.	Listings, quality
harvest_date	Date	Expected harvest or availability date.	Listings
market_location	String/Geo	Market center, district, or GPS coordinate.	Forecasts, listings, orders
forecast_value	Decimal	Predicted demand or price value.	Forecasts
forecast_confidence	Decimal	Model confidence or reliability score.	Forecasts
advisory_label	Enum	SELL SOON, DELAY HARVEST, or NO CHANGE.	Forecast alerts
order_status	Enum	Pending, matched, accepted, dispatched, delivered, settled, disputed.	Orders
agreement_id	String	Identifier for smart agreement or transaction state.	Marketplace
vehicle_id	String	Identifier for registered transport vehicle.	Logistics
vehicle_type	Enum	Reefer, covered truck, open truck, or other type.	Logistics
route_distance_km	Decimal	Estimated route distance.	Logistics
temperature_c	Decimal	Telemetry temperature in degrees Celsius.	Telemetry
humidity_percent	Decimal	Telemetry humidity percentage.	Telemetry
image_path	String	Path or reference to quality grading image.	Quality
spot_severity	Enum	None, mild, or severe visible spot category.	Quality

reputation_score	Decimal	Computed trust score or evidence summary for a farmer.	Marketplace
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Source: Prepared by the FreshRoute project team.

Appendix J: Evaluation metric formulas

The metrics below provide a shared evaluation vocabulary for the four components. Final pilot reporting should state the dataset, sample size, time period, and assumptions used for each metric.

Table J.1: Evaluation metric formulas and interpretation.

Metric	Formula or meaning	Interpretation	Module
MAE	mean absolute error	Average absolute difference between predicted and actual values.	Forecasting
RMSE	root mean squared error	Square root of mean squared prediction error; penalizes large errors.	Forecasting
MAPE	mean absolute percentage error	Average absolute percentage error; unstable when actual values are near zero.	Forecasting
WAPE	weighted absolute percentage error	Total absolute error divided by total actual value.	Forecasting
Match success rate	successful matches / total orders	Share of buyer orders receiving an acceptable fulfillment plan.	Marketplace
Payment delay	settlement timestamp - delivery confirmation timestamp	Time between verified delivery and settlement.	Marketplace
Spoilage rule violation rate	unsafe assignments / total assignments	Share of logistics assignments violating prototype perishability rules.	Logistics
Fulfillment rate	fulfilled orders / total generated or real orders	Share of orders assigned and completed under constraints.	Logistics
F1-score	$2 \times \text{precision} \times \text{recall} / (\text{precision} + \text{recall})$	Class-balanced grading performance indicator.	Quality
Task completion rate	completed user tasks / attempted tasks	Usability success measure for UAT.	Integrated platform

Source: Prepared by the FreshRoute project team.

These appendices should be treated as living implementation material. When FreshRoute moves into a field pilot, each table can be used as a checklist for tracking requirements, services, tests, risks, user workflows, data fields, and metrics.

Appendix K: Deployment readiness checklist

Before FreshRoute is moved from academic prototype to a live pilot, the team should confirm that technical, operational, legal, and support requirements are ready. The checklist below is intentionally practical and should be reviewed with supervisors, pilot partners, and administrators.

Table K.1: Deployment readiness checklist.

Area	Checklist item	Reason
Environment	Separate development, staging, and pilot environments	Reduces risk of test data affecting pilot operations.
Configuration	Secrets stored in environment variables or a secure vault	Prevents accidental exposure of API keys and credentials.
Database	Schema migrations, backups, and restore testing completed	Protects user, order, and evidence records.
Models	Forecast and quality model versions tagged with training metadata	Improves reproducibility and auditability.
APIs	Endpoint documentation and sample payloads available	Supports frontend, testing, and partner integration.
Security	Role-based access, TLS, and authorization checks verified	Prevents unauthorized access to sensitive data.
Payments	Payment gateway sandbox and callback validation tested	Prevents incorrect settlement status and payment disputes.
Telemetry	Sensor gateway tested with realistic route conditions	Ensures alerts are not only simulated in code.
Support	User training guide and support contact prepared	Improves adoption during pilot.

Monitoring	Error logging and usage monitoring enabled	Helps detect failures quickly.
Governance	Dispute resolution and data access policy documented	Protects users and clarifies responsibilities.
Exit plan	Pilot rollback and data export procedure prepared	Reduces risk if pilot must be paused.

Source: Prepared by the FreshRoute project team.

Appendix L: Sample notification templates

FreshRoute uses notifications to turn data into action. Message templates should be short, local-language ready, and clear about whether they are forecasts, confirmations, warnings, or reminders. The following samples can be translated and refined during user testing.

Table L.1: Sample notification templates.

Message type	Sample message	Primary recipient
Forecast alert	Banana demand is expected to rise in Dambulla this week. Consider preparing harvest within 2 days. Confidence: medium.	Farmer
Price drop warning	Mango price trend is falling. Delay harvest if fruit maturity allows. Check FreshRoute for details.	Farmer
Order match	Your 300 kg mango listing matched with Buyer B-104. Review price and pickup time before accepting.	Farmer
Buyer confirmation	Your papaya order has a proposed fulfillment group of 4 farmers. Review reputation and confirm agreement.	Buyer
Vehicle assigned	Trip TR-208 assigned. Pickup: Akuressa 08:30. Destination: Matara Town. Vehicle type: reefer.	Transporter
Temperature alert	Warning: Trip TR-208 temperature exceeded safe limit for 12 minutes. Check vehicle and update status.	Transporter/Admin
Quality result	Mango image graded as Grade B. Main visible reason: mild spots and mixed color stage.	Farmer/Buyer
Settlement notice	Delivery verified for Agreement AG-502. Settlement status updated to processing.	Farmer/Buyer
Complaint update	Complaint CP-09 reviewed. Decision recorded with quality and telemetry evidence.	Buyer/Farmer

Source: Prepared by the FreshRoute project team.

The notification language should avoid technical jargon where possible. Terms such as confidence, grade, and settlement should be explained in the application help section so that users understand what action is expected.

Appendix M: Thesis formatting checklist

This checklist was added to align the group thesis with the SLIIT dissertation structure used as the formatting reference. It can be used before final submission to confirm that layout and content components have been checked.

Table M.1: Thesis formatting checklist.

Section	Requirement checked	Status
Title page	Project title, student names, IDs, degree, department, institution, country, and month/year included.	Completed
Declaration	Candidate and supervisor declaration sections included.	Completed
Abstract	Concise summary of problem, methodology, results, conclusion, and keywords.	Completed
Acknowledgement	Supervisor, co-supervisor, faculty, stakeholders, team, and family support acknowledged.	Completed
Preliminary lists	Table of contents, list of figures, list of tables, abbreviations, and appendices included.	Completed

Formatting	Times New Roman, appropriate heading sizes, 1.5 line spacing for body, and required margins applied.	Completed
Chapters	Introduction, methodology, results and discussion, and conclusion included.	Completed
Commercialization	Product value, rollout, revenue model, and social impact discussed.	Completed
Testing	Module-level and integrated testing strategy included.	Completed
References	IEEE-style reference list included.	Completed
Glossary	Key technical and domain terms defined.	Completed
Appendices	Requirements, APIs, tests, budget, risks, workflows, metrics, and checklists included.	Completed

Source: Prepared by the FreshRoute project team based on SLIIT formatting guidance [33].